

# **Qin Jiushao, Shuxue Jiuzhang, fascicles 1-9**

English Translation

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

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# 数书九章

Mathematical Treatise in Nine Sections

## **Fascicle I – The Dayan Category**

Bilingual Edition / 双语版

By **Qin Jiushao** (秦九韶)

c. 1202–1261 CE

Presented in the 7th year of Chunyou (淳祐七年), 1247 CE

Left side: Original Classical Chinese text

Right side: English translation with modern mathematical commentary

Contents

## 1 Introduction / 导言

### 数书九章序

数书九章者，南宋秦九韶之所著也。秦氏字道古，鲁郡人。生于宋季，博学多能，尤精算学。此书成于淳祐七年（公元1247年），凡九卷，每卷

每问有问、答、术、草四目，体例谨严。大衍之类，尤推精妙，盖中国剩余定理之滥觞也。

#### 1.1 The Nine Categories / 九类名目

九类之目：

1. **大衍** – 求一之术
2. **天时** – 天文历法
3. **田域** – 田地测量
4. **测望** – 勾股重差
5. **赋役** – 租税劳役
6. **钱谷** – 钱粮贸易
7. **营建** – 建筑工事
8. **军旅** – 兵家之事
9. **市易** – 市场交易

### Preface to the Shuxue Jiuzhang

The Shuxue Jiuzhang (Mathematical Treatise in Nine Chapters) is a significant work by the Song Dynasty mathematician Qin Jiushao (秦九韶, c. 1202–1261). Completed in 1247 CE during the Chunyou era (淳祐七年), it stands as one of the greatest achievements in the history of Chinese mathematics and indeed of world mathematics.

The work is organized into nine categories (类), each containing problems (问), for a total of eighty-one problems. Each problem follows a rigorous four-part structure: **Wen** (问) – the problem statement; **Da** (答) – the numerical answer; **Shu** (术) – the general method; **Cao** (草) – the detailed working.

The nine categories are:

1. **Dayan** (大衍) – “Great Derivation” : Indeterminate analysis and the Chinese Remainder Theorem
2. **Tianshi** (天时) – Astronomical phenomena and calendar calculations
3. **Tianyu** (田域) – Land measurement and area calculation
4. **Cewang** (测望) – Surveying and observation
5. **Fuyi** (赋役) – Taxation and labor service
6. **Qian’ gu** (钱谷) – Currency and grain commerce
7. **Yingjian** (营建) – Construction and engineering
8. **Junlv** (军旅) – Military affairs
9. **Shiyi** (市易) – Market transactions

## 2 Preface / 序

周教六艺，数实成之。学士大夫所从事尚矣。其用本太虚生一，而周流无穷。

Of the Six Arts taught by the Zhou, mathematics completes them. It has long been pursued by scholars and gentlemen. Its use originates from the One born of the Great Void, flowing endlessly. In the large, it can penetrate the divine and accord with nature; in the small, it can govern worldly affairs and classify all things. How could it be viewed as shallow or trivial?

若昔推策以迎日，定律而和气。髀距濬川，土圭度晷。天地之大，圉焉而不能外，况其间总悉者乎？爰自河图洛书，闡发幽秘，八卦九畴

From ancient times, reckoning rods were used to predict the sun's course, and laws were established to harmonize the qi. The gnomon measured the rivers, the jade tablet measured the shadows. The greatness of heaven and earth is encompassed by it and cannot escape, how much more the multitude of things within?

极而至于大衍、皇极之用，而人事之变无不该，鬼神之情莫能隐矣。圣人神之言而遗其粗，常人昧之由而莫之觉。要其归，则数与道非二

Reaching to the applications of the Dayan and the Supreme Ultimate, there is no human affair that it does not encompass, no spirit's nature that it cannot reveal. The sages found its spirit and left its roughness; ordinary people are blind to its path. In essence, number and the Way are not two different roots.

独大衍法不载九章，未有能推之者。历家演法颇用之，以为方程者，误也。

Only the Dayan method is not recorded in the Nine Chapters, and none have been able to develop it. Calendarists have used it extensively in their computational methods, but those who take it as a system of equations are mistaken.

九韶愚陋，不闲于艺。然早岁侍亲中都，因得访习于太史，又尝从隐君子受数学。时陈兵难，历岁遯塞，不自意至于矢石之间。更险离忧

I Jiushao am foolish and unrefined, not skilled in the arts. Yet in my youth I served my parents in the capital, and so had the opportunity to study with the Grand Astrologer; I also received instruction in mathematics from a gentleman recluse. When the time of military disaster came, I did not expect to survive between arrows and stones. Passing through danger and sorrow, ten years elapsed; my heart withered and my spirit fell. Truly I came to know that all things have number.

时淳祐七年九月，鲁郡秦九韶叙。

Written in the ninth month of the seventh year of Chunyou, by Qin Jiushao of Lu Commandery.

### 3 The Dayan General Method / 大衍总数术

#### 3.1 Editor’ s Commentary / 按语

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 按大衍术，以各分数之奇零求各分数之总数。大而天行，小而物数，皆可御之。 | Editor’ s note: The Dayan method takes the remainders (odd fragments) of each fractional divisor to find the total number. Whether for large celestial motions or small material quantities, it can be applied. The method has the following steps: finding the original numbers, finding the definitive numbers, finding the derivative numbers, finding the odd numbers, finding the multipliers, and finding the use numbers. |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 3.2 The Four Types of Numbers / 四种问数

大衍总数术曰：置诸问数，类名有四。

- 1. 元数 – 谓尾位见单零者
- 2. 收数 – 谓尾位见分厘者
- 3. 通数 – 谓诸数各有分子分母者
- 4. 复数 – 谓尾位见十或百及千以上者

**The Dayan General Method states:** Place the given numbers; there are four types.

- 1. **Original numbers** – Numbers ending in single units
- 2. **Collected numbers** – Numbers ending in fractional parts
- 3. **Common numbers** – Numbers each having numerator and denominator
- 4. **Repeated numbers** – Numbers ending in tens, hundreds, or thousands

#### 3.3 Finding the Definitive Numbers / 求定数

元数者：先以两两连环求等，约奇弗约偶。或约得五而彼有十，乃约偶弗约奇，或无数偶偶，约之可存一位见偶。

**For original numbers:** First, pairwise in chain sequence reduce the odd, not the even one. If reducing yields 5 while the other has 10, then reduce the even one, not the odd one. If all original numbers are even, after reduction one may keep one even number.

复数者：问数尾位见十以上者，以诸数求总等，存一位约众位，始得元数。两两连环求等，约奇弗约偶，复乘偶，或约偶弗约奇。

**For repeated numbers:** When the given numbers end in tens or above, find the GCDs; reduce the odd, not the even, and multiply back the even; or reduce the even, not the odd, and do not multiply back the odd.

#### 3.4 Computing the Derivative Numbers / 求衍数

求定数勿使两位见偶，勿使见一太多。见一多则借用繁，不欲借则任得偶。

以定相乘为衍母，以各定约衍母，得各衍数。

次以各定母满去衍数，各余名曰奇数。

以大衍求一术，得各乘率。

以乘衍数，各得用数。

**Finding definitive numbers:** Do not let two positions be even; do not let there be too many ones. Too many ones makes borrowing complicated.

**Multiply the definitive numbers together to obtain the derivative mother.** Reduce the derivative mother by each definitive number to obtain each **derivative number**. Next, reduce each derivative number by its corresponding definitive number; the remainders are called the **odd numbers**.

Using the **Dayan Method for Finding Unity**, obtain each **multiplier**.

Multiplying each multiplier by its derivative number gives each **use number**.

#### 3.5 The Dayan Method for Finding Unity / 大衍求一术

|                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 大衍求一术云：置奇右上，定居右下。立天元一于左上。先以右行上下两位，以少除多，所得商数，乃递互乘归左行。须使右上末后奇一归一，方止。 | The Dayan Method for Finding Unity says: Place the odd number at the upper right, the definitive number at the lower right. Place the celestial element unity (1) at the upper left. First, divide the larger by the smaller of the two numbers in the right column; the quotient obtained is then used to multiply alternately into the left column. Continue until the upper right finally becomes one. Then check what is obtained at the upper left—this is the multiplier. |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

In modern mathematical terms: Given coprime integers  $a$  (the “odd number” ) and  $m$  (the “definitive num-

ber” ), find  $k$  such that:

$$k \cdot a \equiv 1 \pmod{m}$$

This is the **modular multiplicative inverse** of  $a$  modulo  $m$ , computed by the **extended Euclidean algorithm**.

4 Problem 1: Shigua Fawei / 蓍卦发微

| 蓍卦发微 – Divination by Yarrow Stalks                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p><b>问曰：</b><br/>《易》曰：大衍之数五十，其用四十有九。又曰：分而为二以象两，挂十以象一，揲十以象四，挂八以象三，挂六以象二。问：何谓大衍之数五十，其用四十有九？</p>                            | <p><b>Question:</b><br/>The I Ching says: “Divide, of the Great Derivation is fifty; of these, forty-nine are used.” It also says: “Divide into two to symbolize the Two [heaven and earth]; hang one to symbolize the Three [heaven, earth, and man]; count off by fours to symbolize the Four Seasons.” Three changes produce a line; eighteen changes produce a hexagram. The problem asks to find the computational method of this derivation and the numbers involved. This problem applies the Dayan method to the classical yarrow-stalk divination of the I Ching.</p>                                                                                                                                                                 |
| <p><b>答曰：</b><br/>衍母十二，衍法三。<br/>一元衍数二十四，二元衍数一十二，三元衍数八，四元衍数六。以上四位衍数，共五十一。<br/>一揲用数一十二，二揲用数二十四，三揲用数四，四揲用数九。以上四位用数，共四十九。</p> | <p><b>Answer:</b><br/>The derivative mother is 12, the derivative divisor is 3. The first derivative number is 24, the second is 12, the third is 8, the fourth is 6. The sum of these four derivative numbers is 51.<br/>The first manipulation use-number is 12, the second is 24, the third is 4, the fourth is 9. The sum of these four use-numbers is 49.<sup>a</sup></p>                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>术曰：</b><br/>置诸元数，两两连环求等，约奇弗约偶。遍约毕，乃变元数皆曰定母，列在行各立天元一为子，列在右；以诸定母互乘左行之子，各得名曰衍数。</p>                                    | <p><b>Method:</b><br/>Place the given original numbers in pairs; divide the odd and not the even. When reduction is complete everywhere, the original numbers become the definitive numbers; place them in the right column. Place the celestial element unity (1) as the seed in the left column. Cross-multiply the definitive numbers with the seeds in the left column; each result is called a derivative number.<br/>Next, reduce each derivative number by its corresponding definitive number; each remainder is called the odd number. Using the Dayan Method for Finding Unity on the odd number and the definitive number, obtain each multiplier. Multiply each derivative number by its multiplier to obtain each use number.</p> |
| <p>次以各定母满去衍数，各余名曰奇数。以奇数与定母用大衍术求一，得各乘率以乘衍数，各得用数。</p>                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



## 5 Problem 2: Guli Huiji / 古历会积

### 古历会积 – Ancient Calendar Accumulation

问曰：

问古历会积，欲求积年。以历法积元为问，历过无考，但据近世演纪之法，推而求之。

Question:

This problem concerns the calculation of the accumulated years (积年) in ancient calendar systems. Given certain astronomical periodicities, one must find a number that simultaneously satisfies multiple congruence conditions. The epochs have passed without record; based on the method of computing epochs used in recent times, one extrapolates backward.

This was one of the primary applications of the Dayan method in traditional Chinese astronomy—finding a common epoch that aligns multiple celestial cycles.

术曰：

以大衍求之。置元历相关诸周期，连环求等，约奇弗约偶，得定母。

Method:

诸定相乘为衍母。以各定约衍母得衍数。各满足母去 period 余为奇数。大衍求原历，两两在链序列中找 GCDs，约奇弗约偶，获得定数。将定数相乘得衍母。将衍母除以各定数得衍数。将衍数乘以各定数得用数。将用数相加得总数。将总数除以衍母，余数即为所求积年。

6 Problem 3: Tuiji Tugong / 推计土功

推计土功 – Estimating Earthworks

|                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p><b>问曰：</b><br/>四县（甲、乙、丙、丁）共同筑堤，根据各县的物力分配筑堤任务。</p> <ul style="list-style-type: none"><li>甲县物力：138,600贯</li><li>乙县物力：146,300贯</li><li>丙县物力：192,500贯</li><li>丁县物力：184,800贯</li></ul> <p>筑堤标准：每770贯物力分配一名劳动力，每人每天筑堤60平方尺。<br/>进度情况：甲县和乙县已完成任务，丙县剩余51丈，丁县剩余18丈。</p> | <p><b>Question:</b><br/>Four counties (A, B, C, D) jointly build a dike; the task is distributed according to each county' s resources.</p> <ul style="list-style-type: none"><li>County A resources: 138,600 guan</li><li>County B resources: 146,300 guan</li><li>County C resources: 192,500 guan</li><li>County D resources: 184,800 guan</li></ul> <p>Dike standard: for every 770 guan of resources, assign one laborer; each laborer builds 60 square feet of dike per day.<br/>Progress: Counties A and B have completed their tasks; County C has 51 zhang remaining; County D has 18 zhang remaining.<br/>This is a classic indeterminate problem solved by the Dayan method.</p>                                                                                    |
| <p><b>答曰：</b><br/>堤的总长度为19里235步5尺。<br/>各县所筑长度：</p> <ul style="list-style-type: none"><li>甲县：1,260丈</li><li>乙县：1,768步5尺6寸</li><li>丙县：4里328步5尺6寸</li><li>丁县：与前三县相应</li></ul>                                                                                        | <p><b>Answer:</b><br/>The total length of the dike is 19 li 235 bu 5 chi.<br/>Each county' s contribution:</p> <ul style="list-style-type: none"><li>County A: 1,260 zhang</li><li>County B: 1,768 bu 5 chi 6 cun</li><li>County C: 4 li 328 bu 5 chi 6 cun</li><li>County D: corresponding to the first three</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>术曰：</b><br/>以大衍求之。置各县物力，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。</p>                                                                                                                                                                                   | <p><b>Method:</b><br/>Solve by the Dayan Method. Place the resources of each county; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each county' s remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.</p> |

约奇非的偶，得定母，诸定相乘为行母，以定约行母得衍数。各满定母去

treasury A; subtracting the decrement each time gives each treasury's rate. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

将 1085 乘以 1085 的奇数的偶数，得到定母。诸定相乘为衍母 (in sheng): the official measure 83 sheng, the Anji measure 110 sheng, the Pingjiang measure 135 sheng. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

相乘为母。The Dayan Method for Finding the GCDs of all the odd numbers in a chain sequence. 大衍求一各满足母之奇数大衍求一  
travel distances; pairwise in chain sequence find GCDs,  
reducing the odd and not the even, obtaining the definitive  
numbers. Multiply the definitive numbers together to  
obtain the derivative mother. Reduce the derivative mother  
by each definitive number to obtain the derivative numbers.  
Reduce each by its definitive number; the remainders are  
the odd numbers. Apply the Dayan Method for Finding  
Unity to obtain the multipliers. Multiply the derivative  
numbers by the multipliers to obtain the use numbers.  
Next, place each remainder, multiply by the use numbers,  
and sum to obtain the total number. Reduce by the  
derivative mother; the remainder is the sought distance.

10 Problem 7: Chengxiang Xiangji / 程行相及

| 程行相及 – Catching Up on a Journey                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p><b>问曰：</b><br/>甲、乙、丙三人行程相关。甲先行若干日，乙后行，丙又后行。各以不同速度行进，求某处，三人相遇之时间及各地距离。</p>       | <p><b>Question:</b><br/>Three travelers, A, B, and C, start on different journeys. A departs first, B departs later, and C departs still later. Each travels at a different speed. Find: the time when all three meet at a certain place, and the various distances. This problem involves multiple travelers with different speeds and starting times, requiring the Dayan method to solve a system of congruences.</p>                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>术曰：</b><br/>以大衍求之。置各人行程速度及时间差，连环求等，约奇弗约偶，得定母。诸定母相乘，以各定约行母得衍数。各定母去之，余为奇数。</p> | <p><b>Method:</b><br/>Solve by the Dayan method. Place each person's speed and time difference; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.</p> |

11 Problem 8: Jichi Xunyuan / 积尺寻源

| 积尺寻源 – Finding Dimensions from Volume                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <p><b>问曰：</b><br/>问欲砌基一段，见管大小方砖六门城砖四色。令匠取便，或平或侧，只用一色砖砌，须要适足。匠以破量地计料，并用</p> <ul style="list-style-type: none"><li>大方：广多六寸，深少六寸</li><li>小方：广多二寸，深少三寸</li><li>城砖：长广多三寸，深少一寸</li><li>六门砖：长广多三寸，深多一寸</li></ul> <p>皆不合匝，未免修破砖料裨补。其四色砖尺寸：</p> <ul style="list-style-type: none"><li>大方：方一尺三寸</li><li>小方：方一尺一寸</li><li>城砖：长一尺二寸，阔六寸，厚二寸五分</li><li>六门砖：长一尺，阔五寸，厚二寸</li></ul> <p>欲知基深广几何？</p> <p><b>答曰：</b><br/>深三丈七尺一寸，广一丈二尺三寸。</p> <p><b>术曰：</b><br/>以大衍求之。置四色砖尺寸及所余尺寸，连环求等，约奇弗约偶，得定</p> | <p><b>Question:</b><br/>One mason, to build a foundation, has four types of bricks—large and small square bricks, six-portal bricks, and city-wall bricks—four types in all. The mason is instructed to use whichever is convenient, laying them flat or on edge, using only one type of brick, requiring them to fit exactly. The mason measured the ground and estimated materials, reporting:</p> <ul style="list-style-type: none"><li>Large square: width excess 6 cun, depth deficit 6 cun</li><li>Small square: width excess 2 cun, depth deficit 3 cun</li><li>City brick: length and width excess 3 cun, depth deficit 1 cun</li><li>Six-portal: length and width excess 3 cun, depth excess 1 cun</li></ul> <p>All fail to fit exactly, so broken bricks must be used for patching. The four types of brick have dimensions:</p> <ul style="list-style-type: none"><li>Large square: 1 chi 3 cun square</li><li>Small square: 1 chi 1 cun square</li><li>City brick: 1 chi 2 cun long, 6 cun wide, 2 cun 5 fen thick</li><li>Six-portal: 1 chi long, 5 cun wide, 2 cun thick</li></ul> <p>Find: the depth and width of the foundation.<br/>This is a classic problem of finding a common multiple with given remainders.</p> <p><b>Answer:</b><br/>Depth: 3 zhang 7 chi 1 cun; width: 1 zhang 2 chi 3 cun.</p> <p><b>Method:</b><br/>Solve by the Dayan Method. Place the four types of brick dimensions and the excess/deficit dimensions; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder gives the depth and width of the foundation.</p> |

12 Problem 9: Yumi Tuishu / 余米推数

余米推数 – Calculating Rice Remainders

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>问曰：</b><br/>有米若干，以不同容量的容器量之，各余若干。欲求米之总数。</p>    | <p><b>Question:</b><br/>There is a quantity of rice; when measured with containers of different capacities, there are different remainders each time. Find the total amount of rice.<br/>This is the classic “unknown quantity” problem. In modern terms: find <math>N</math> such that <math>N \equiv r_i \pmod{m_i}</math> for given moduli <math>m_i</math> and remainders <math>r_i</math>.</p> |
| <p><b>术曰：</b><br/>以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。</p> | <p><b>Method:</b><br/>以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。此即大衍求一术。以各定约母得衍数，各满定母去之，余为奇数。大衍求一术，即今之中国剩余定理。以各奇数乘衍数，相加，得定母，即得米之总数。</p>                                                                                                                                                                                                                                                                        |

This problem is structurally identical to the famous “problem of the unknown quantity” from the Sunzi Suanjing (孙子算经), which asks: “We have things of which we do not know the number; if we count them by threes, the remainder is 2; if by fives, the remainder is 3; if by sevens, the remainder is 2. How many things are there?” Qin Jiushao’s general method solves all problems of this type systematically.



### 13 Mathematical Analysis / 数理分析

#### 13.1 The Chinese Remainder Theorem / 中国剩余定理

The Dayan method solves systems of simultaneous linear congruences. Given a set of pairwise coprime moduli  $m_1, m_2, \dots, m_n$  and remainders  $r_1, r_2, \dots, r_n$ , find  $N$  such that:

$$N \equiv r_1 \pmod{m_1}$$
$$N \equiv r_2 \pmod{m_2}$$
$$\vdots$$
$$N \equiv r_n \pmod{m_n}$$

#### 13.2 Qin Jiushao’s Algorithm / 秦九韶算法

- 求定数 – 连环求等，约奇弗约偶
- 求衍母 – 诸定相乘
- 求衍数 – 以定约衍母
- 求奇数 – 各满定母去之
- 求乘率 – 大衍求一术
- 求用数 – 乘率乘衍数
- 求总数 – 余数乘用数并之

- Step 1: Definitive Numbers. Given the original numbers (元数), reduce them pairwise using the GCD, keeping them coprime. The rule “reduce the odd, not the even” ensures unique factorization.
- Step 2: Derivative Mother. Compute  $M = m_1 \times m_2 \times \dots \times m_n$ .
- Step 3: Derivative Numbers. Compute  $M_i = M/m_i$  for each  $i$ .
- Step 4: Odd Numbers. Compute  $g_i = M_i \bmod m_i$ .
- Step 5: Finding Unity. For each  $i$ , find  $k_i$  such that  $k_i \cdot g_i \equiv 1 \pmod{m_i}$ . This is the modular multiplicative inverse, computed by the extended Euclidean algorithm.
- Step 6: Use Numbers. Compute  $u_i = k_i \cdot M_i$ .
- Step 7: Final Computation. The solution is:  $N = \sum_{i=1}^n r_i \cdot u_i \pmod{M}$

#### 13.3 Historical Significance / 历史意义

秦氏大衍术，超前欧洲数百年。其求一之术，即欧几里得之扩展算法也。后世郭守敬用之于天文，李冶用于几何，在西传欧洲，遂成所谓“中国剩余定理”。高斯于《算术探究》（1801年）首次在欧洲完整论述此定理，称之为“求一术”（以备已知数除之，各余同知数）问题。今此定理为数论之基本算法。

Qin Jiushao’s method predated similar European work by several centuries. It was used by Guo Shoujing in astronomy, Li Ye in geometry, and later in Europe, becoming known as the “Chinese Remainder Theorem.” Gauss gave the first complete treatment in Europe in his *Disquisitiones Arithmeticae* (1801), calling it the “problem of finding a number that, when divided by given numbers, leaves given remainders.” The theorem is now recognized as one of the fundamental results in number theory.

## About This Edition / 关于本版

This bilingual edition presents Fascicle I of Qin Jiushao’s *Shuxue Jiuzhang* (Mathematical Treatise in Nine Sections), with the original classical Chinese text in the **left column** and an English translation with modern mathematical commentary in the **right column**.

### Technical Notes / 技术说明

- Typeset with XeTeX using parallel `minipage` environments
- Font: Noto Sans CJK SC (supports both Chinese and Latin characters)
- The traditional structure of 问 (question), 答 (answer), 术 (method), and 草 (working) has been preserved
- Translator’s footnotes explain technical terminology and historical context

### Translation Notes / 翻译说明

- “Dayan” (大衍) — translated as “Great Derivation,” preserving the I Ching reference
- “Qiuyi” (求一) — translated as “Finding Unity” (finding the modular inverse)
- “Yanshu” (衍数) — translated as “derivative numbers”
- “Yanmu” (衍母) — translated as “derivative mother”
- “Chengshu” (乘率) — translated as “multiplier”
- “Yongshu” (用数) — translated as “use number”

### Further Reading / 参考文献

- Libbrecht, U. *Chinese Mathematics in the Thirteenth Century*. MIT Press, 1973.
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# MATHEMATICAL TREATISE IN NINE SECTIONS

Shuxue Jiuzhang – 數學九章

## Fascicle 2

### 天時類 – Tian Shi Lei

*Heavenly Patterns: Calendar, Astronomy and Weather*

Bilingual Edition / 雙語版

Original Chinese

原文

English Translation

英譯

Qin Jiushao / 秦九韶 (1202–1261)

Song Dynasty, Chunyou 7th year (淳祐七年, 1247)

*Parallel bilingual edition with original Chinese and English translation  
based on the Siku Quanshu (四庫全書) edition*

“數術之傳，以實為體。” / “The art of numbers, rooted in the Great Void...”

**Contents**

## Introduction / 導言

### 【原文】

數學九章卷二天時類，凡九問。上卷五問：推氣治歷、治歷推閏、治歷演紀、綴術推星、揆日究微，皆關乎曆法推步之事。下卷四問：天池測雨、圓罍測雨、峻積驗雪、竹器驗雪，皆量雨驗雪之法。

秦九韶自序云：「數術之傳，以實為體。其大可以通神明、順性命；其小可以經世務、類萬物。詎容以淺近窺哉？」

此卷九問，示數學之實用於曆法、天文、氣象之間，於農桑之國尤為切要。

### [English Translation]

The Tian Shi Lei (天時類, “Heavenly Patterns”) is the second of nine chapters in Qin Jiushao’s *Shuxue Jiuzhang* (數學九章, *Mathematical Treatise in Nine Sections*). This fascicle contains **nine problems** divided into two parts:

- Calendar computation and astronomical predictions (Problems 1–5)
- Measurement of rainfall and snowfall (Problems 6–9)

Qin Jiushao’s preface to the *Shuxue Jiuzhang* (1247) explains his motivation:

*“The art of numbers, rooted in the Great Void, generates One and circulates without end. In its large aspect it can penetrate the divine and comply with destiny; in its small aspect it can govern worldly affairs and classify the myriad things. How can it be viewed as shallow or narrow?”*

The nine problems of this fascicle demonstrate the practical application of mathematical techniques to calendar reform, planetary motion, and meteorological measurement—topics of vital importance to agricultural society.

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# 1 Volume 2, Part 1 / 卷二上

## 1.1 Problem 1 / 第一問: 推氣治歷

### 【原文】 推氣治歷

#### 問

太史觀天，測景驗氣：

- 慶元四年戊午歲（1198 年）冬至，三十九日九十二刻四十五分。
- 紹定三年庚寅歲（1230 年）冬至，三十二日九十四刻十二分。

今欲求嘉泰甲子歲（1204 年）氣骨、歲餘、斗分各若干。

#### 按

按：紹定三年庚寅冬至，實為紹定四年辛卯歲首。辛卯距戊午三十四年，為積三十三年。

#### 答曰

- 氣骨：一十一日三十八刻二十分八十一秒八十小分。
- 歲餘：五日二十四刻二十九分三十秒三十小分。
- 斗分：零日二十四刻二十九分三十秒三十小分。

#### 術曰

以兩測相距之年數為法。置前測之日刻分，以減後測之日刻分；不減，加紀策。累加紀策，至合天道為止。以五日上為實。實如法得歲餘。去全日，餘為斗分。以歲餘乘前測至所求中間之年數，加前測日刻分，滿紀策去之；餘為所求氣骨。

### [English] Calculating Qi and Regulating the Calendar

#### Question

The Imperial Astronomer observed the heavenly way:

- In the 4th year of Qingyuan (慶元四年, 1198, year Wu-wu 戊午), the winter solstice was at 39 days, 92 *ke* (刻), 45 *fen* (分).
- In the 3rd year of Shaoding (紹定三年, 1230, year Geng-yin 庚寅), the winter solstice was at 32 days, 94 *ke*, 12 *fen*.

It is desired to find, for the intermediate year Jiatai Jiazi (嘉泰甲子, 1204): the *qi bone* (氣骨), the *year remainder* (歲餘), and the *dou fraction* (斗分).

#### Editor's Note

The Shaoding 3rd year Geng-yin winter solstice actually marks the beginning of Shaoding 4th year Xin-mao (辛卯). Xin-mao is 34 years from Wu-wu, giving 33 accumulated years.

#### Answer

- Qi bone: 11 days, 38 *ke*, 20 *fen*, 81 *miao* (秒), 80 small fractions.
- Year remainder: 5 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.
- Dou fraction: 0 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.

#### Method

First take the number of years between the two observations as the divisor. Place the earlier observation's days, *ke*, and *fen*, and subtract from the later observation's days, *ke*, and *fen*; the remainder is the rate. (If insufficient for subtraction, add the *ji-ce* 紀策). Accumulate by adding *ji-ce* repeatedly until the result is suitable for the heavenly way—using a number above 5 days as the dividend. Divide the dividend by the divisor to obtain the year remainder. Remove whole days; the remainder is the *dou fraction*. Multiply the year remainder by the number of years from the earlier observation to the desired intermediate year, add to the earlier observation's days, *ke*, and *fen*, and remove full *ji-ce*; the remainder is the desired *qi bone*.

**按**

氣骨者，冬至後夜半甲子之日分也。歲餘者，歲周減去六甲之餘也。斗分者，歲周減去三百六十五日之餘也。此法尚未知歲實之確數，故以兩測氣骨之差為實、積年為法，推而求之。

**Editor's Note**

The qi bone is the days and fractions from the winter solstice to the Jiazi day after midnight. The year remainder is the residue when the solar year length is reduced by six Jiazi cycles. The dou fraction is the residue when the solar year is reduced by 365 days. This method does not yet know the precise solar year length, hence it uses the difference between two observed qi bones as the dividend and accumulated years as divisor.

## 1.2 Problem 2 / 第二問: 治歷推閏

## 【原文】 治歷推閏

## 問

開禧曆，以嘉泰四年甲子歲：

- 天正冬至：一十一日乙亥四十四刻六十一分五十四秒。
- 十一月經朔：一日乙丑七十五刻五十五分六十二秒。

問：閏骨、閏率各幾何？

## 答曰

- 閏骨：九日六十九刻五分九十一秒，餘一百六十一分之一百二十一。
- 閏骨率：一十六萬三千七百七十一。

## 術曰

以日法通氣朔日刻分秒，各為氣骨分、朔骨分。氣骨分以約率約之，恰盡則可用；或不盡而餘不及一刻，亦可用。乃以減朔骨分；餘為閏骨率。以日法除之，得閏骨策。

## 按

若以朔骨分徑減冬至分，餘為九日六十九刻五分九十二秒，即閏骨策。原草僅差一百六十一分之四十八。其不徑減而用通分約分、輾轉乘除者，為後續計算之便也。

## [English] Regulating the Calendar and Intercalary Months

## Question

For the Kaixi calendar (開禧曆), taking the 4th year of Jiatai (嘉泰四年, 1204, year Jiazi 甲子):

- The celestial winter solstice is at 11 days, day-token Yi-hai (乙亥), 44 *ke*, 61 *fen*, 54 *miao*.
- The 11th-month new moon (經朔) is at 1 day, day-token Yi-chou (乙丑), 75 *ke*, 55 *fen*, 62 *miao*.

Question: What are the *run bone* (閏骨) and *run rate* (閏率)?

## Answer

- Run bone: 9 days, 69 *ke*, 5 *fen*, 91 *miao*, with remainder 121/(169 fractions).
- Run bone rate: 163,771.

## Method

Take the day-factor (日法) to convert the *qi* and new-moon days, *ke*, *fen*, and *miao* into *qi-bone* and new-moon-bone fractions respectively. The *qi-bone* fraction, when reduced by the approximation rate (約率), if exactly divisible, is usable; or if the remainder after rounding is below one *ke*, it is also usable. Then subtract from the new-moon-bone fraction; the remainder is the *run bone* rate. Divide by the day-factor to obtain the *run bone ce* (策).

## Editor's Note

If one directly subtracts the new moon fraction from the winter solstice fraction, the remainder is 9 days, 69 *ke*, 5 *fen*, 92 *miao*, giving the *run bone ce*. The original working only differs by 48/(169 fractions). The reason for not directly subtracting but instead using common denominators, approximation, and repeated multiplication/division is to facilitate subsequent calculations.



## 1.3 Problem 3 / 第三問: 治歷演紀

## 【原文】 治歷演紀

## 問

開禧曆積年七百八十四萬八千一百八十三。欲知其演紀之源：調日法、求朔餘、朔率、斗分、歲率、歲閏、入元歲、入閏、朔定骨、閏泛骨、閏縮、紀率、氣元率、元閏、元數、氣等率、因率、部率、朔等數、因數、部數、朔積年，凡二十三事。問各幾何？

## 答曰

(選錄要項：)

- 日法：一萬六千九百
- 朔餘：八千九百六十七
- 朔率：四十九萬九千六十七
- 斗分：二十四刻二十九分三十秒三十小分
- 歲率：六百一十七萬二千六百八
- 歲閏：二萬七百二十八
- 入元歲：九千二百四十
- 入閏：三萬一千六十八

## 術曰

以大衍術求氣分、朔分、紀分及衍母、諸正用。以衍母除氣分，得積年；除朔分，得積月；除紀分，得積紀。以六十乘之，得積日。以氣骨乘紀正用得紀總；以閏骨乘朔正用得朔總。併二總，滿衍母去之，餘為所求率實。以氣分除之，得歷過年；以減積年，得未至年。

## 按

此問盡顯秦九韶大衍求一術之威力，以解同餘式組，用於曆法推算。所求二十三事，構成自天文觀測推導開禧曆上元之完整體系。

## [English] Deriving Calendar Epoch Calculations

## Question

The Kaixi calendar has an accumulated year count of 7,848,183. It is desired to know the derivation of this computation: adjusting the day-factor, finding the new-moon remainder, new-moon rate, dou fraction, year rate, year intercalation, entry-year, entry-intercalation, new-moon fixed bone, intercalation floating bone, intercalation shrinkage, ji rate, qi yuan rate, yuan intercalation, yuan number, qi equal-rate, cause-rate, *bu* rate, new-moon equal-number, cause-number, *bu* number, and new-moon accumulated year—23 items in total. What is each?

## Answer

(Selected key results:)

- Day-factor (日法): 16,900
- New-moon remainder (朔餘): 8,967
- New-moon rate (朔率): 499,067
- Dou fraction (斗分): equivalent to 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions
- Year rate (歲率): 6,172,608
- Year intercalation (歲閏): 20,728
- Entry-year (入元歲): 9,240
- Entry-intercalation (入閏): 31,068

## Method

Use the method of Dayan (大衍術) to find the qi fraction, new-moon fraction, ji fraction, and derived mother (衍母), and all the proper use-numbers (正用). Divide the derived mother by the qi fraction to obtain the accumulated years; by the new-moon fraction to obtain the accumulated months; by the ji fraction to obtain the accumulated ji. Multiply by 60 to obtain accumulated days. Multiply the qi bone by the ji proper-use to obtain the ji total; multiply the run bone by the new-moon proper-use to obtain the new-moon total. Combine the two totals, remove full derived mothers, and the remainder is the sought rate-dividend. Divide by the qi fraction to obtain the calendar-passed (歷過) years; subtract from the accumulated years to obtain the years not yet reached.

## Editor's Note

This problem demonstrates the full power of Qin Jiushao's Dayan method (大衍求一術) for solving systems of congruences, applied here to calendar computation. The 23 quantities computed form a complete system for deriving the epoch parameters of the Kaixi calendar from astronomical observations.

## 1.4 Problem 4 / 第四問: 綴術推星

## 【原文】 綴術推星

## 問

歲星合伏段：一十六日九十分，行三度九十分，去日一十三度乃見。然後順行一百一十三日，一十七度八十三分，乃留。欲知：

- 合伏段常度、初行率、末行率、平行率
- 晨疾初段常度、初行率、末行率

各幾何？

## 術曰曰（綴術）

1. 合伏段：常度即總日數。初行率即總行度數。末行率以初行率減日差乘（日數減一）。
2. 晨疾初段：常度即所給日數。初行率即前段末行率。末行率以初行率減日差乘（日數減一）。
3. 平行率：並初末二行率而半之，得平行率。

## 按

五星之行，有遲疾不等。此法僅取合伏、見段之中率，而以累加累減求逐日之行。古法之疏，於五星尤甚。原文多隱奧難明，下詳解之，以見古今精粗之比。

## 要項計算：

- 合伏段：常度一十六日九十分；初行率二十二分七秒；末行率二十一分九十六秒；平行率約二十二分。
- 晨疾初段：常度三十日；初行率二十一分九十六秒；日差一十一秒二十三分六十六小秒。

## [English] Calculating Planetary Motion

## Question

The year-star (Jupiter, 歲星) in conjunction-hidden phase (合伏段): after 16 days 90 *fen*, it moves 3 degrees 90 *fen*, removing 13 degrees from the sun before becoming visible. Then it proceeds directly (順行) for 113 days, 17 degrees 83 *fen*, before halting (乃留). It is desired to know:

- The conjunction-hidden segment (合伏段) constant degree, initial motion-rate, final motion-rate, mean motion-rate
- The morning-rapid initial segment (晨疾初段) constant degree, initial motion-rate, final motion-rate

What is each?

## Method

(Method of Zhui Shu):

1. For the conjunction-hidden segment: The constant degree is the total number of days. The initial motion rate is the total degrees moved. The final motion rate equals the initial rate minus the daily difference (日差) multiplied by (days – 1).
2. For the morning-rapid initial segment: The constant degree is the given number of days. The initial rate is the final rate of the previous segment. The final rate equals the initial rate minus the daily difference multiplied by (days – 1).
3. For the mean motion rate: Add the initial and final rates and halve; this gives the parallel (mean) rate.

## Editor's Note

The five planets' motions involve non-uniform acceleration and deceleration. The method here takes only the middle values of the conjunction-hidden and direct-visibility segments, then uses successive addition/subtraction to derive the daily motion. The ancient method's approximation is particularly evident for the five planets. The original text's language was mostly hidden and obscure; the following detailed explanation reveals the comparison between ancient and modern precision.

## Key calculations:

- Conjunction-hidden segment: Constant degree 16 days 90 *fen*; Initial rate 22 *fen* 7 *miao*; Final rate 21 *fen* 96 *miao*; Mean rate approx. 22 *fen*.
- Morning-rapid initial segment: Constant degree 30 days; Initial rate 21 *fen* 96 *miao*; Daily difference 11 *miao* 23 small-*fen* 66 small-*miao*.

## 1.5 Problem 5 / 第五問: 揆日究微

## 【原文】 揆日究微

## 問

歷代測景，唯唐大衍曆最密。本朝崇天曆測陽城：

- 冬至影：一丈二尺七寸一分五十秒
- 夏至影：一尺四寸七分七十秒

與大衍曆合。今開禧曆測臨安府：

- 冬至影：一丈八寸二分二十五秒
- 夏至影：九寸一分

欲知：臨安夏至後幾日，其影與陽城夏至影等？與大衍曆影比較，相差幾何？

## 答曰

大暑後五日正午，影長一尺四寸八分八十五秒（半）。影差六寸一分二十九秒（少）。

## 術曰曰

以乘法乘之，節率入焉。

1. 置臨安所測冬至影，減夏至影；餘為景差，以為實。
2. 置象限度（九十一度三十一分四十四秒），加一十一度二十五分二十七秒半；以度為寸，得一百零二寸五千六百七十一分五十秒，以為法。
3. 實如法得零點九六六四……；棄餘，自乘得節泛數：九千四百分。
4. 以諸節氣乘節率，加夏至影幕，開方得諸節氣影長。

## [English] Measuring the Sun's Shadow

## Question

Among all historical shadow-measurements, only the Tang dynasty's Dayan calendar (大衍曆) was the most precise. This dynasty's Chongtian calendar (崇天曆) gives for Yangcheng (陽城):

- Winter solstice shadow: 1 *zhang* 2 *chi* 7 *cun* 1 *fen* 50 *miao*
- Summer solstice shadow: 1 *chi* 4 *cun* 7 *fen* 70 *miao*

These agree with the Dayan calendar. Now the Kaixi calendar (開禧曆) gives for Lin'an (臨安府):

- Winter solstice shadow: 1 *zhang* 8 *cun* 2 *fen* 25 *miao*
- Summer solstice shadow: 9 *cun* 1 *fen*

It is desired to find: after how many days past the summer solstice at Lin'an will the shadow equal the Yangcheng summer solstice shadow? And comparing with the Dayan calendar's shadow, by how much do they differ?

## Answer

Five days after the Great Heat (大暑後五日) at noon; the shadow length is 1 *chi* 4 *cun* 8 *fen* 85 *miao* (half). The shadow difference is 6 *cun* 1 *fen* 29 *miao* (less).

## Method

Use the multiplication-rate method, with the section-rate (節率) entering into it.

1. Place Lin'an's measured winter solstice shadow, subtract the summer solstice shadow; the remainder is the shadow-difference (景差), taken as dividend.
2. Place the 象限度 (91 degrees 31 *fen* 44 *miao*), add 11 degrees 25 *fen* 27.5 small-*fen*; treating degrees as *cun*, obtain 102 *cun* 5671.5 *fen* 50 *miao* as divisor.
3. Divide the dividend by the divisor, obtaining 0.9664...; discard the remainder and square to obtain the section floating-number (節泛數): 9,340 *fen*.
4. Multiply by the section multiplication-rates for each solar term, add the summer solstice shadow-squared, take the square root to obtain each solar term's shadow length.

**按**

各節氣影長，當時實測而定，本不須推算。此術故為繁難，以惑學者。細考之：象限度加一十一度有餘以為法；以景差除之而自乘，得節率。每節氣有乘率；以節率乘之，加夏至影幂，得各節氣之影幂。故節率者，強立之數也。

**Editor's Note**

Each solar term's shadow length was determined by actual measurement at the time and does not need computation. The method presented here deliberately creates obscurity to puzzle the reader. Detailed examination shows: the 象限度 plus 11+ degrees is taken as divisor; the shadow-difference divided by this and then squared gives the section-rate. Each solar term has a multiplication-rate; multiplying the section-rate by this and adding the summer solstice shadow-power gives each term's shadow-power. Thus the section-rate is an artificially extracted number.

## 2 Volume 2, Part 2 / 卷二下

## 2.1 Problem 6 / 第六問: 天池測雨

## 【原文】 天池測雨

## 問

今州郡皆有天池盆以測雨水。然世人但知盆中之水即所得之雨，而不知器形不同則受雨各異 函 故不可徑以盆測為平地之雨數。

設盆口徑二尺八寸，底徑一尺二寸，深一尺八寸。接雨水深九寸。問平地雨深幾何？

## 答曰

平地雨深：三寸。

## 術曰曰（圓錐術）

1. 以口徑乘底徑，又各自乘；併三數。以半深乘之；又以十一乘、七除，得下積。
2. 以半深加雨深，以減全深，得上深。
3. 以底徑減口徑；以餘乘上深。以半深乘之，加口徑乘半深，得面率。
4. 以面率除以半深，得水表面徑。
5. 續以乘除，求得雨水體積，以口冪（平方三倍）除之，得平地雨深。

## 按

數學提要：天池盆為圓錐之截頭體。所積雨水體積為

$$V_{\text{雨}} = \frac{\pi h_{\text{雨}}}{3} (R^2 + Rr + r^2)$$

以口冪  $\pi R^2$  除之，得平地雨深。

## [English] Measuring Rain with the Heavenly Pool Basin

## Question

Nowadays every prefecture and district has a “heavenly pool” basin (天池盆) for measuring rainwater. But people only know that the water in the basin equals the rain received, and do not understand that different vessel shapes receive different amounts of rain—so one cannot directly take the basin measurement as the flat-ground rain amount.

Suppose the basin has:

- Mouth diameter (口徑): 2 *chi* 8 *cun* (28 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)
- Depth (深): 1 *chi* 8 *cun* (18 *cun*)
- Rainwater depth inside (接雨水深): 9 *cun*

What is the equivalent flat-ground rain depth?

## Answer

Flat-ground rain depth: 3 *cun*.

## Method (Use the method for circular cones / 圓錐術):

1. Multiply the mouth diameter by the base diameter, and also square each; combine these three values. Multiply by half the depth; then multiply by 11 and divide by 7 to obtain the lower volume (下積).
2. Take half the depth plus the rain depth, subtract from the total depth to obtain the upper depth (上深).
3. Subtract the base diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth, add the result to the mouth-diameter times half-depth, to obtain the surface rate (面率).
4. Divide the surface rate by half the depth to obtain the water surface diameter.
5. Continue with further multiplications to isolate the rain volume, then divide by the mouth-area (squared and tripled) to obtain the flat-ground rain depth.

## Editor's Note

Mathematical summary: The heavenly pool basin is a frustum of a cone. The rain volume collected is:

$$V_{\text{rain}} = \frac{\pi h_{\text{rain}}}{3} (R^2 + Rr + r^2)$$

where  $R$  is the top radius,  $r$  is the base radius, and  $h_{\text{rain}}$  is the rain depth. Dividing by the mouth area  $\pi R^2$  gives the equivalent flat-ground rain depth.

## 2.2 Problem 7 / 第七問: 圓罌測雨

## 【原文】 圓罌測雨

## 問

以圓罌受雨：口徑一尺五分（一尺零五分），腹徑二尺四寸，底徑八寸，深一尺六寸。接雨水深一尺二寸。以密率（ $\pi \approx 22/7$ ）入圓術。問平地雨深幾何？

## 答曰

平地雨深：一尺八寸加  $\frac{6483}{74088}$  寸。（校正本）

## 術曰

以底徑乘腹徑，又各自乘；併三數。以器半深乘之；以十一乘、四十二除，得下積。

以器半深加雨深，以減全深，得上深。以腹徑減口徑；以餘乘上深。以半深乘之，加口徑乘半深，得面率。以面率乘腹率；又各自乘併之；以十一乘、四十二除，得上率。併二率（上下）為雨水總積實。以口徑平方三倍除之，得平地雨深。

## 按

原題云「以密率入圓術」，於求平地雨深無涉。若求器之雨水體積，則此語為當。原草有誤，上呈校正法。

## [English] Measuring Rain with a Round Vessel

## Question

Using a round vessel (圓罌) to catch rain:

- Mouth diameter (口徑): 1 *chi* 5 *fen* (10.5 *cun*)
- Belly diameter (腹徑): 2 *chi* 4 *cun* (24 *cun*)
- Base diameter (底徑): 8 *cun*
- Depth (深): 1 *chi* 6 *cun* (16 *cun*)
- Rainwater depth inside: 1 *chi* 2 *cun* (12 *cun*)

Using the “dense rate” (密率,  $\pi \approx 22/7$ ) for circle computations, what is the flat-ground rain depth?

## Answer

Flat-ground rain depth: 1 *chi* 8 *cun* plus  $\frac{6483}{74088}$  *cun*. (Editor’s corrected value.)

## Method

Multiply the base diameter by the belly diameter, and also square each; combine these three values. Multiply by half the vessel depth; multiply by 11 and divide by 42 to obtain the lower volume.

Take half the vessel depth plus the rain depth, subtract from the total depth to obtain the upper depth. Subtract the belly diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth; add the mouth-diameter times half-depth to obtain the surface rate.

Multiply the surface rate by the belly rate; also square each and combine; multiply by 11 and divide by 42 to obtain the upper rate. Combine the two rates (upper and lower) to obtain the total rain volume as dividend. Divide by three times the squared mouth-diameter to obtain the flat-ground rain depth.

## Editor’s Note

The original problem statement mentions “dense rate for circle computation” which is irrelevant to finding the flat-ground rain depth. If one were seeking the vessel’s rain volume, then this phrase would be appropriate. There were errors in the original computation; the above presents the corrected method.



## 2.3 Problem 8 / 第八問: 峻積驗雪

## 【原文】 峻積驗雪

## 問

驗雪以占年：牆高一丈二尺。以板倚之，去址五尺，板末與牆頭齊。板上積雪厚四寸。峻處積薄而平處積厚。問平地雪深幾何？

## 答曰

平地雪深：一尺四分。

## 術曰曰（少廣術，連枝法）

1. 以去址自乘，為隅。
2. 以牆高自乘；併入上隅。
3. 以雪厚自乘；乘上併數，為實。
4. （可約則約之。）開連枝平方，得平地雪深。

## 草曰

皆以寸為單位：

- 去址五十寸；自乘得二千五百（隅）。
- 牆高一百二十寸；自乘得一萬四千四百。
- 併之：一萬四千四百加二千五百 = 一萬六千九百。
- 雪厚四寸；自乘得一十六。
- 相乘：16900 × 16 = 270400（實）。

隅與實同約百；得實二千七百零四、隅二十五。開平方： $\sqrt{2704/25} = 52/5 = 10.4$  寸 = 一尺四分。

## 按

理與術皆正。實即勾股術，雖不以此名之。「連枝」之號，隱其理也。圖示：AB 為平地雪深（與牆頂雪厚等），BC 為板上餘雪。三角形 ABC 與板倚牆之形相似。牆高為大股，去址為大勾，板為大弦。以勾股定理，比例求之，得平地雪深。

## [English] Verifying Snow Accumulation

## Question

To verify snow for predicting the harvest: A wall is 1 zhang 2 chi (12 chi) high. A plank leans against it, the base 5 chi from the wall's foot, with the tip level with the wall top. Snow accumulates 4 cun thick on the plank. The steep accumulation is thin while the flat accumulation is thick. What is the flat-ground snow depth?

## Answer

Flat-ground snow depth: 1 chi 4 fen.

**Method** (Use the method of Shao Guang / 少廣, with the connected-branch / 連枝 method):

1. Square the distance from the base (去址): this is the corner (隅).
2. Square the wall height; combine with the corner above.
3. Square the snow thickness; multiply by the combined value above to obtain the dividend.
4. (Reduce if possible.) Extract the connected-branch square root to obtain the flat-ground snow depth.

## Working

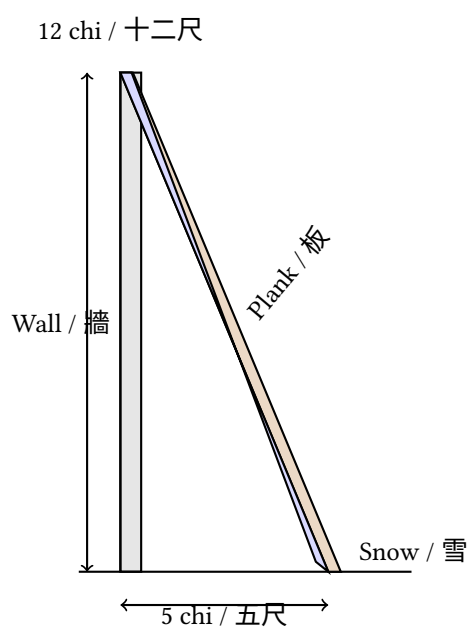
Convert all measurements to cun:

- Distance from base: 50 cun; squared = 2,500 (corner).
- Wall height: 120 cun; squared = 14,400.
- Combined: 14,400 + 2,500 = 16,900.
- Snow thickness: 4 cun; squared = 16.
- Multiply: 16,900 × 16 = 270,400 (dividend).

The corner and dividend share a factor of 100; reduce to get dividend 2,704 and corner 25. Extracting the square root:  $\sqrt{2704/25} = 52/5 = 10.4$  cun = 1 chi 4 fen.

## Editor's Note

Both the principle and method are correct. In fact, this uses the right-triangle (勾股) method, though not named as such. The “connected-branch” terminology conceals the underlying principle. The diagram illustrates: AB is the flat-ground snow depth (equal to the wall-top snow depth), BC is the excess snow on the plank. Triangle ABC is similar to the triangle formed by the plank leaning against the wall. The wall height is the large leg, the distance from base is the large base, the plank is the large hypotenuse. The flat-ground snow depth is found by proportional reasoning via the Pythagorean theorem.





## 2.4 Problem 9 / 第九問: 竹器驗雪

## 【原文】 竹器驗雪

## 問

以圓竹器驗雪：口徑一尺六寸，深一尺七寸，底徑一尺二寸。雪入其中，深一尺。器通透風，雪入既多，則平地之數淺。問平地雪厚幾何？

## 答曰

平地雪厚：九寸加  $\frac{764}{3439}$  寸。

## 術曰

以底徑減口徑；以餘乘雪深，半之，自乘，為隅。以器深自乘，乘雪深自乘；併入隅，乘雪深自乘，為實。開連枝三乘方，得平地雪厚。

## 草曰

皆以寸為單位：

- 口徑十六寸；底徑十二寸。
- 差四寸；雪深十寸。
- 相乘： $4 \times 10 = 40$ ；半之得二十（上廉）。
- 器深自乘： $17^2 = 289$ 。
- 雪深自乘： $10^2 = 100$ 。
- 相乘： $289 \times 100 = 28900$ ；併入隅，續開三乘方。

其草用「三乘方開法」，得平地雪厚九寸，餘  $\frac{764}{3439}$ 。

## 按

「通透風」之語，於術無涉。或謂器中之雪深因風而異於平地；此問實與天池測雨同理，以截頭圓錐形之竹器而變通之。

## [English] Verifying Snow with a Bamboo Container

## Question

Using a round bamboo container (圓竹器) to verify snow:

- Mouth diameter (口徑): 1 *chi* 6 *cun* (16 *cun*)
- Depth (深): 1 *chi* 7 *cun* (17 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)

Snow falls into it, filling to a depth of 1 *chi*. The container is permeable to wind; when much snow enters, the flat-ground depth is less. What is the flat-ground snow thickness?

## Answer

Flat-ground snow thickness: 9 *cun* plus  $\frac{764}{3439}$  *cun*.

## Method

Subtract the base diameter from the mouth diameter; multiply the remainder by the snow depth, halve, and square to obtain the corner (隅). Square the container depth and multiply by the squared snow depth; combine with the corner and multiply by the squared snow depth to obtain the dividend. Extract the connected-branch cube root (三乘方) to obtain the flat-ground snow thickness.

## Working

Convert all measurements to *cun*:

- Mouth diameter: 16 *cun*; base diameter: 12 *cun*.
- Difference: 4 *cun*; snow depth: 10 *cun*.
- Multiply:  $4 \times 10 = 40$ ; halve = 20 (this is the upper-lian / 上廉).
- Container depth squared:  $17^2 = 289$ .
- Snow depth squared:  $10^2 = 100$ .
- Multiply:  $289 \times 100 = 28,900$ ; combine with the corner and continue the cube-root extraction.

The working proceeds through the Chinese “method of extracting cube roots with carrying” (三乘方開法), yielding the flat-ground snow depth of 9 *cun* with remainder 764/3439.

## Editor’s Note

The phrase “permeable to wind” does not affect the computation method. Some commentators suggest the snow depth in the container differs from the flat-ground depth due to wind effects; this problem uses the same geometric principle as the Heavenly Pool rain problem, adapted for a frustum-shaped bamboo container.

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| 2   | 治歷推閏       | Zhi Li Tui Run    | Finding intercalation parameters for a calendar / 求閏骨閏率               |
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End of Fascicle 2 / 卷二終

「七曜周旋於蒼穹，人事追綴以相通。  
星以夜候，晷以晝察 幽歷年既久，術愈疏；  
唯智者能革而新之。」

— 秦九韶《數學九章》自序

*“Seven planets revolve in the firmament; the affairs of men are tracked by pursuing and connecting [their motions].  
Stars by night, gnomon by day—as ages pass, [methods] become coarse;  
only the wise can reform them.”*

— Qin Jiushao, Preface to *Shuxue Jiuzhang*

SIKU QUANSHU EDITION / 四庫全書本

# Shu Xue Jiu Zhang / 數學九章

Fascicles 3–4 / 卷三至四

Tian Yu (田域) + Ce Wang (測望)

**Author:** Song Dynasty, Qin Jiushao / 宋 秦九韶

**Dates:** c. 1202–1261

**Topics:** Juan 3 Shang/Xia: Tian Yu (Fields / 田域類)

Juan 4 Shang/Xia: Ce Wang (Surveying / 測望類)

**Bilingual Edition / 雙語版**

LEFT: Original Chinese / 原文 | RIGHT: English Translation / 英譯



*Typeset from the Siku Quanshu (四庫全書) edition*

Bilingual edition with parallel Chinese text and English translation

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## Introduction / 緒言

數學九章為南宋秦九韶所撰，是中古數學之偉大著作。全書九章，每章一類。

本卷呈其第三、四卷：

- 卷三上：田域類 – 方田、圓田、三角田及組合圖形之面積
- 卷三下：田域類續 – 續田域問題，含籌算圖式
- 卷四上：測望類 – 隔河測高遠，用勾股重差之法
- 卷四下：測望類續 – 續測望問題，量不可至之遠近高下

文本依中算傳統格式：

- 問 – 問題陳述
- 答 – 答數
- 術 – 演算方法
- 草 – 詳細演草

The *Shu Xue Jiu Zhang* (Mathematical Treatise in Nine Chapters), written by Qin Jiushao during the Southern Song Dynasty, is one of the greatest masterpieces of medieval Chinese mathematics. The work contains nine chapters, each devoted to a different class of problems.

This volume presents Fascicles 3 and 4 of the work, covering:

- **Juan 3 Shang – Tian Yu (Fields, Part 1):** Problems on area measurement, including squares, circles, triangles, and composite figures.
- **Juan 3 Xia – Tian Yu (Fields, Part 2):** Continuation of field problems, including more complex configurations with counting rod diagrams.
- **Juan 4 Shang – Ce Wang (Surveying, Part 1):** Problems on remote measurement and surveying, using similar triangles and the gou-gu method.
- **Juan 4 Xia – Ce Wang (Surveying, Part 2):** Further surveying problems, including measurement of heights, distances, and inaccessible objects.

The text follows the classical format of Chinese mathematical literature:

- **Wen (問):** The problem statement
- **Da Yue (答):** The answer
- **Shu Yue (術):** The method/algorithm
- **Cao Yue (草):** The working/calculation

# 1 Juan 3 Shang – Tian Yu / 卷三上田域類

小序：此卷論田域面積之術。涉方圓、斜直、面羣、體積，皆相求之。其本則出於立天元一法。

*Introductory note:* This fascicle treats problems on area measurement (Tian Yu). The problems involve squares, circles, triangles, and composite figures. Qin Jiushao employs his “method of the celestial element” (Tian Yuan Shu) alongside traditional algorithmic techniques.

## Problem 1: Fang Zhong Gu Yuan / 方中古圓池

問/ 第一問：方中古圓池

有方中古圓池，因陂毀於一角。從外方隅斜至內圓邊七尺六寸。欲依舊修之，求圓徑、方面、方斜各幾何。

答/ 答曰：

池圓徑三丈六尺六寸，四百二十九分寸之四百一十二。

方面三丈六尺六寸，四百二十九分寸之四百一十二。

方斜五丈一尺八寸，四百二十九分寸之四百一十二。

術/ 術曰：

以少廣求之，太求入之。如積斜自乘倍之為實，倍斜為一方法，一半寸為從隅。開平方得徑。

## Problem / Square with Inscribed Circular Pond

There is a square field with an ancient circular pond inscribed within it. The northern corner has partially collapsed. From the corner of the outer square to the edge of the inner circle, the diagonal distance is 7 chi 6 cun (7.6 chi). Wishing to repair it according to the original design, find: the diameter of the circle, the side of the square, and the diagonal of the square.

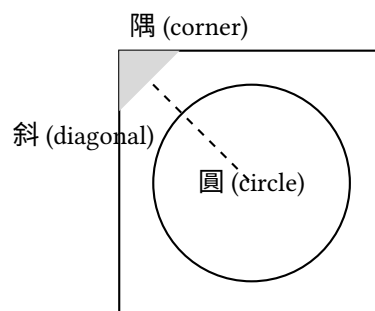
**Answer / Answer:**

The pond's diameter is 36 chi 6 cun plus  $\frac{412}{429}$  cun; the square's side is 36 chi 6 cun plus  $\frac{412}{429}$  cun; the square's diagonal is 51 chi 8 cun plus  $\frac{412}{429}$  cun.

**Method / Method:**

Using the “lesser breadth” (shao guang) method with the “transformation of the embryo” (tai tai) technique. Square the diagonal and double it to obtain the dividend (shi). Take twice the diagonal as the negative coefficient of the linear term (yi fang). Take half a cun as the coefficient of the quadratic term (cong yu). Extract the square root to find the diameter.

Tu (Diagram / 圖):



## Problem 2: San Xie Qiu Ji / 三斜求積

問/ 第二問：三斜求積

有三斜形田，大斜一十三里，小斜一十四里，中斜一十五里。里法三百步，欲知為田積幾何。

## Problem / Triangular Field Area

There is a triangular field. The large diagonal (da xie) is 13 li, the small diagonal (xiao xie) is 14 li, and

答/ 答曰：

田積三百一十五頃。

術/ 術曰：

以少廣求之。以小斜幂並大斜幂，減中斜幂，餘半之，自乘於上。以小斜幂乘大斜幂，減上，餘四約之為實。一為從隅，開平方得積。

the middle diagonal (zhong xie) is 15 li. With 300 bu per li, find the area.

**Answer / Answer:**

The area is 315 qing.

**Method / Method:**

Using the “lesser breadth” method. Add the squares of the small and large diagonals, subtract the square of the middle diagonal. Halve the remainder and square it (placed above). Multiply the square of the small diagonal by the square of the large diagonal; subtract the result above; divide the remainder by 4 to get the dividend. Take 1 as the coefficient of the quadratic term; extract the square root to find the area.

*In modern notation:* If  $a = 13$ ,  $b = 14$ ,  $c = 15$ , then:

$$S = \sqrt{\frac{1}{4} \left[ a^2 b^2 - \left( \frac{a^2 + b^2 - c^2}{2} \right)^2 \right]}$$

This is equivalent to **Heron’s formula**:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

### Problem 3: Fang Tian Yuan Chi / 方田圓池

問/ 第三問：方田圓池

有方田一段，內有圓池水占之，外計地三千一百六十八步。只云從內池周至外田角斜二十一步半。內外方圓各幾何。

答/ 答曰：

內池周四十二步，徑一十三步半。方田面六十步。

術/ 術曰：

以少廣求之。置積以圓率三約之為實，倍斜為從方，斜幂為一負隅。開平方得內徑。加倍斜得方面。

### Problem / Square Field with Central Circular Pond

There is a square field with a circular pond inside it. The remaining land outside the pond amounts to 3168 bu. It is given that the diagonal distance from the edge of the inner pond to the corner of the outer square is 21.5 bu. Find the dimensions of both the inner circle and outer square.

**Answer / Answer:**

The inner pond’s circumference is 42 bu, its diameter is 13.5 bu; the square field’s side is 60 bu.

**Method / Method:**

Using the “lesser breadth” method. Place the area and divide by the circular rate 3 to obtain the dividend. Twice the diagonal gives the linear coefficient. The square of the diagonal gives the negative quadratic coefficient. Extract the square root to find the inner diameter. Add twice the diagonal to obtain the square’s side.

### Problem 4: Piao Tian / 漂田

問/ 第四問：漂田

有漂田一段，中斜一十六步，水止五步。減中斜一十六餘一十一為元大斜。內減殘大斜二十步……

### Problem / The Drowned Field

There is a “drowned field” (a triangular field partially submerged). The middle diagonal is 16 bu; the water depth is 5 bu. Subtracting 5 from 16 gives 11 bu as the

答/ 答曰：

田積若干。

術/ 術曰：

草曰：以水止五減中斜一十六餘一十一步為法。以中斜一十六乘大殘二十得三百二十為大斜實。以法除之得二十九步一十一分步之一為元大斜。內減殘大斜二十步，餘九步一十一分步之一……

base large diagonal.

**Answer / Answer:**

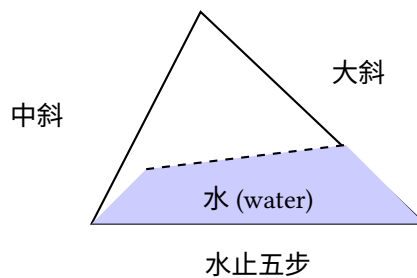
The field area is [to be determined].

**Method / Method:**

(Detailed calculation with specific numerical operations as recorded in the original text.)

*Working:* Take the water depth 5 subtracted from the middle diagonal 16, remainder 11 bu as the divisor. Multiply the middle diagonal 16 by the large remainder 20 to get 320 as the large diagonal dividend. Divide by the divisor to get 29 and  $\frac{1}{11}$  bu as the original large diagonal. Subtract the large remainder 20 bu, remainder 9 and  $\frac{1}{11}$  bu...

Tu (Diagram / 圖):





## 2 Juan 3 Xia – Tian Yu Xu / 卷三下田域類續

### Problem 5: San Hu Fen Tian / 三戶分田

問/ 第五問：三戶分田

有田一段，欲分與甲乙丙三戶。甲多一百五十步，乙多一百二十步，丙多九十步。三戶共田若干，各得幾何。

答/ 答曰：

甲田若干，乙田若干，丙田若干。

術/ 術曰：

以少廣求之。併三戶多出之數以減總積，餘以三戶約之，各得本積。又各加多出之數即得各戶之田。

**Problem / Division of Land Among Three Families**

There is a field to be divided among three families jia, yi, and bing. Family jia receives 150 bu more than the base amount; family yi receives 120 bu more; family bing receives 90 bu more. If the total area is given, find each family's share.

**Answer / Answer:**

Family jia's share [is determined]; family yi's share [is determined]; family bing's share [is determined].

**Method / Method:**

Using the "lesser breadth" method. Add the excess amounts from all three families and subtract from the total area. Divide the remainder by 3 to get the base amount for each. Add each family's excess to the base amount to obtain each share.

### Problem 6: Yuan Tian Fang Tian / 圓田方田

問/ 第六問：圓田方田

有方圓田各一段，共積一千二百八十步。只云方面如圓徑少四步，問方圓面徑各幾何。

答/ 答曰：

方面若干，圓徑若干。

術/ 術曰：

以少廣求之。置共積以圓率三、方率四各約之為實，方面少圓徑四步為從方。開平方得圓徑。減四得方面。

**Problem / Circular and Square Fields**

There is a square field and a circular field. Their total area is 1280 bu. It is given that the square's side is 4 bu less than the circle's diameter. Find the side and diameter.

**Answer / Answer:**

The square's side [and] the circle's diameter [are determined].

**Method / Method:**

Using the "lesser breadth" method. Place the combined area; divide by the circular rate 3 and square rate 4 to obtain the dividend. The difference of 4 bu between square side and circle diameter gives the linear coefficient. Extract the square root to find the diameter. Subtract 4 to obtain the square's side.

### Problem 7: Hu Tian / 弧田

問/ 第七問：弧田

有弧田一段，弦五十步，矢二十五步，求積若干。

答/ 答曰：

田積若干。

術/ 術曰：

以弦矢求之。以弦乘矢，又以矢自乘，併之，半之為實。開平方得積。

**Problem / Circular Segment Field**

There is a circular segment field (hu tian) with chord (xian) 50 bu and arrow (shi) 25 bu. Find its area.

**Answer / Answer:**

The field area [is determined].

**Method / Method:**

Using the chord-arrow method. Multiply the chord by the arrow; add the square of the arrow; halve to obtain the dividend. Extract the square root to find the area.

*Note:* For a circular segment with chord  $c$  and sagitta  $s$ , Qin's formula gives:

$$A = \frac{cs + s^2}{2}$$

This is an approximation to the true area of a circular segment.

Counting Rod Diagram / 籌算圖式:

| 弦    | 矢  | 弦乘矢   | 矢幂             |
|------|----|-------|----------------|
| 50   | 25 | 1250  | 625            |
| 併    |    | 半之    | 積              |
| 1875 |    | 937.5 | $\approx 30.6$ |

### 3 Juan 4 Shang – Ce Wang / 卷四上測望類

小序：此卷論測望之術，量不可至之高遠。用勾股、重差之法，為中國古代測量學之基礎。

*Introductory note:* This fascicle treats problems on surveying and remote measurement (Ce Wang). These problems involve measuring inaccessible distances and heights using the “hook-and-gourd” (gou gu) method and similar triangles.

#### Problem 1: Yuan Cheng / 圓城

問/ 第一問：圓城

有圓城不知周徑，四門中開。北外三里有喬木，出南門折而東行九里乃見木。欲知城周徑各幾何。

答/ 答曰：

徑九里，周二十七里。

術/ 術曰：

以勾股夕桀求之。一為從隅，五因北里為從七廉，置北里冪八因為從五廉，以北里乘上廉為一從，三連負隅乘五廉為一上廉，以北里乘上廉為實。開玲瓏九乘方得數，自乘為徑。以三因徑得周。

#### Problem / The Round City

There is a circular city of unknown circumference and diameter. It has four gates at the cardinal points. Three li north of the north gate stands a tall tree. Going out the south gate and then turning east for 9 li, the tree becomes visible. Find the circumference and diameter of the city.

**Answer / Answer:**

The diameter is 9 li, the circumference is 27 li.

**Method / Method:**

Using the “hook-and-gourd evening-peak” method. Take 1 as the leading coefficient. Multiply the north distance by 5 for the seventh-degree coefficient. Place the square of the north distance, multiply by 8 for the fifth-degree coefficient. Multiply the north distance by the upper coefficient for the negative linear coefficient. Double the negative rate to form the fifth-degree coefficient as the negative upper coefficient. Multiply the north distance by the upper coefficient for the dividend. Extract the ninth-degree root (ling long kai fang) to obtain the number; square it for the diameter. Multiply the diameter by 3 for the circumference.

#### Problem 2: Ce Ta / 測塔

問/ 第二問：測塔

有塔高未知其數，去塔若干步立一表，人目上望見塔尖，退行若干步又立一表，人目上望仍見塔尖。求塔高及去塔遠近。

答/ 答曰：

塔高若干，去塔遠若干。

術/ 術曰：

以勾股求之。兩表退行之差為法，前表去塔遠乘表高，後表去塔遠乘表高，相減為實。實如法而一得塔高。又以法除前表去塔遠乘兩表高差加表高得塔高。

#### Problem / Measuring a Tower

A tower of unknown height stands at an unknown distance. At a certain distance from the tower, a pole is erected; looking upward from eye level, the tower's peak is visible. Retreating a certain distance and erecting another pole, the peak is again visible from eye level. Find the tower's height and distance.

**Answer / Answer:**

The tower height [and] distance to the tower [are determined].

**Method / Method:**

Using the hook-and-gourd method. The difference between the retreat distances of the two poles is the divisor. Multiply the front pole's distance to the tower by the pole height; multiply the rear pole's distance by the pole height; subtract to obtain the dividend. Divide to

get the tower height. Alternatively, multiply the front distance by the difference of pole heights, divide by the divisor, and add the pole height.

*In modern notation:* Let  $d_1$  = front distance,  $d_2$  = rear distance,  $h$  = pole height. Then:

$$\text{Tower height} = \frac{d_2 \cdot h - d_1 \cdot h}{d_2 - d_1} = h$$

This uses the principle of similar triangles, where the ratio of heights equals the ratio of distances.

### Problem 3: Fang Cheng / 方城

問/ 第三問：方城

有方城一座，不知大小。北門外有樹，出南門直行若干步折而西行若干步見樹。求城方面。

答/ 答曰：

城方面若干步。

術/ 術曰：

以勾股求之。兩行步相乘為實，併兩行步為從方。開平方得城方面。

### Problem / The Square City

There is a square city of unknown size. Outside the north gate stands a tree. Going out the south gate and walking straight a certain number of steps, then turning west and walking a certain number of steps, the tree becomes visible. Find the side of the square city.

**Answer / Answer:**

The city's side is [determined] in bu.

**Method / Method:**

Using the hook-and-gourd method. Multiply the two walk distances for the dividend. Add the two walk distances for the linear coefficient. Extract the square root to obtain the city's side.

If  $a$  = southward steps and  $b$  = westward steps:

$$x^2 + (a+b)x = ab \quad \Rightarrow \quad x = \frac{-(a+b) + \sqrt{(a+b)^2 + 4ab}}{2}$$

(  
1em)

### Problem 4: Chou Suan Kai Fang / 籌算開方

問/ 第四問：籌算開方

問以開方術解高次方程，其籌佈列之法若何。

術/ 術曰：

開方之法，先列實於中，次列隅於下，次列方於上。商除之，以上乘隅生廉，以上乘廉生方，以上乘方除實。實盡則止，不盡則倍方，三廉為方法，續商除之。

### Problem / Counting Rod Root Extraction

On the method of extracting roots to solve higher-degree equations, how are the counting rods arranged?

**Method / Method:**

In the method of root extraction, first place the dividend (shi) in the middle, then place the unit coefficient (yu) below, and the linear coefficient (fang) above. Divide by estimation; multiply the estimate by the unit coefficient to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. When the dividend is exhausted, stop. If not exhausted, double the linear coefficient and proceed to the next digit.

Counting Rod Layout / 籌式:

|                                    |     |     |     |   |
|------------------------------------|-----|-----|-----|---|
| 商                                  | 實   | 方   | 廉   | 隅 |
| 7                                  | 360 | 130 | ... | 1 |
| <i>Iterative extraction</i> / 逐次開方 |     |     |     |   |

*Note:* In the counting rod system, vertical strokes represent units (1, 2, 3, 4) and horizontal strokes represent fives. The symbol T represents 6 (5+1). The diagram shows the iterative process of root extraction.

## 4 Juan 4 Xia – Ce Wang Xu / 卷四下測望類續

### Problem 5: Ce Shui Shen / 測水深

問/ 第五問：測水深

有井不知其深，以繩測之，繩長倍於井深餘三尺；三摺而測之適足。求井深及繩長。

答/ 答曰：

井深三尺，繩長九尺。

術/ 術曰：

以盈不足求之。倍繩餘三尺為盈，三摺適足為適足。以盈乘適足為實，併盈適足為法，除之得井深。倍之加盈得繩長。

### Problem / Measuring Depth of Water

There is a well of unknown depth. Measuring with a rope, the rope's length is twice the well's depth with 3 chi to spare. Folding the rope in three and measuring, it fits exactly. Find the depth of the well and the length of the rope.

**Answer / Answer:**

The well is 3 chi deep; the rope is 9 chi long.

**Method / Method:**

Using the “surplus and deficit” (ying bu zu) method. The 3 chi excess when doubled is the surplus; the exact fit when tripled is the exact amount. Multiply the surplus by the exact amount for the dividend; add surplus and exact amount for the divisor; divide to obtain the well depth. Double and add the excess for the rope length.

*In modern notation:* Let  $d$  = well depth,  $L$  = rope length.

$$L = 2d + 3, \quad \frac{L}{3} = d$$

Substituting:  $2d + 3 = 3d \Rightarrow d = 3, L = 9$ .

### Problem 6: Ce Ta Gao / 測塔高

問/ 第六問：測塔高

有塔高未知，去塔百步立一表高五尺，人目去表一尺八寸，上望塔尖與表端參合。退行六十步，人目去表端仍參合塔尖。求塔高及塔心去表遠近。

答/ 答曰：

塔身高三丈，塔高一十一丈七尺，與八丈七尺為塔心目長。

術/ 術曰：

此皆大小形同勢相求法也。人目去塔為總股，人目去杆為分小股，人目上塔尖高塔身節為總股高。相論高為分大勾，數一杆去塔分大勾於小角杆去塔為分大勾。

### Problem / Measuring the Height of a Pagoda

A pagoda of unknown height stands at an unknown distance. 100 bu from the pagoda, a 5-chi pole is erected. The observer's eye is 1.8 chi from the ground; looking up, the pagoda's peak aligns with the pole's tip. Retreating 60 bu, the observer's eye again aligns with both the pole's tip and the pagoda's peak. Find the pagoda's height and its distance from the pole.

**Answer / Answer:**

The pagoda's body is 3 zhang high; the total height (from ground to peak) is 11 zhang 7 chi; the distance from the pagoda's center to the measuring point is 8 zhang 7 chi.

**Method / Method:**

These are all applications of similar triangles (da xiao xing tong shi). The distance from eye to pagoda is the total height line; the distance from eye to pole is the small height line. The height from eye level to pagoda peak is the total height; the height from eye level to pole top is the small height. The method uses proportional relationships between these quantities.

*This problem uses the principle of similar triangles:*

$$\frac{\text{Pagoda height} - \text{eye height}}{\text{Distance to pagoda}} = \frac{\text{Pole height} - \text{eye height}}{\text{Distance to pole}}$$

(  
1em)

### Problem 7: Ge He Ce Yuan / 隔河測遠

問/ 第七問：隔河測遠

問隔河望一高岸，不知其遠。平地立一表，人目上望見岸頂，又退立一表望之仍見岸頂。求河闊及岸高。

答/ 答曰：  
河闊若干步，岸高若干尺。

術/ 術曰：

以重差求之。兩表退行步數之差為法，前表去岸遠乘表高為實，實如法而一得岸高。又以法除後表去岸遠乘表高亦得岸高，兩者相驗。

(  
1em)

### Problem 8: Jiu Cheng Fang / 九乘方

問/ 第八問：九乘方

問以開九乘方術解方程，其立法之原若何。

術/ 術曰：

開方之術，至於九乘方，其理一也。先列實於中，次定隅位，次立方廉法。商除之際，以上乘隅生廉，以上乘廉生方，以上乘方除實。每進一位，則方退一，廉退二，隅退三。如是疊進，直至實盡為止。

### Problem / Measuring Distance Across a River

Across a river, a high bank is visible at an unknown distance. A pole is erected on level ground; looking up from eye level, the bank's top is visible. Retreating and erecting another pole, the bank's top is again visible. Find the width of the river and the height of the bank.

**Answer / Answer:**

The river width [and] the bank height [are determined].

**Method / Method:**

Using the “double difference” (chong cha) method. The difference between the two retreat distances is the divisor. Multiply the front pole's distance to the bank by the pole height for the dividend; divide to obtain the bank height. Similarly for the rear pole; the two results should agree (providing verification).

### Problem / The Nine-Degree Root Method

On the method of extracting ninth-degree roots to solve equations – what is the fundamental principle of this method?

**Method / Method:**

The method of root extraction, up to the ninth degree, follows the same principle. First place the dividend in the middle; then determine the unit position; then establish the linear and intermediate coefficients. During division-by-estimation: multiply the estimate by the unit coefficient to generate the intermediate coefficients; multiply the estimate by the intermediate coefficients to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. At each decimal shift, the linear coefficient descends one place, the intermediate descend two places, and the unit descends three. Continue iteratively until the dividend is exhausted.

*Note:* This describes **Horner's method** for solving polynomial equations, which Qin Jiushao developed in-





## Appendix / 附錄: Mathematical Notes

### A.1 籌算 / The Counting Rod System

籌算為中國古代主要計算工具。以竹籌佈列十進位制：

- 一至四：縱式 |, ||, |||, ||||
- 五：橫式 ≡
- 六至九：組合，如 T (五加一)
- 零：以空位或圓 ○ 表之  
籌列於算板，每列一位。此位值制助成高深代數運算，如開方、解方程。

### A.1 The Counting Rod System (Chou Suan)

The counting rod system (Chou Suan) was the primary computational tool in ancient Chinese mathematics. Numbers were represented by bamboo sticks arranged in a decimal place-value system:

- **Units (1–4):** Vertical strokes |, ||, |||, ||||
- **Fives:** A horizontal stroke = placed over the units
- **6–9:** Combinations, e.g., T (5+1), ≡ (3 horizontal over vertical)
- **Zero:** Represented by a blank space or circle ○

The rods were arranged on a counting board in columns, with each column representing a decimal place. This positional system enabled sophisticated algebraic computations including root extraction and solution of polynomial equations.

### A.2 天元術 / The Celestial Element Method

秦九韶精於天元術，為早期代數符號。未知數謂之「天元」，方程以籌式佈列。此法可解十次方程，早霍納五百餘年。

### A.2 The “Celestial Element” Method (Tian Yuan Shu)

Qin Jiushao was a master of the “celestial element” method, an early form of algebraic symbolism. The unknown quantity was called “celestial element” (Tian Yuan, tian yuan), and equations were set up using counting rod arrangements. This method could solve polynomial equations of degree up to 10, anticipating Horner’s method by over 500 years.

### A.3 三斜求積公式 / Qin’s Formula for Triangular Area

三角形三邊  $a, b, c$ ，秦九韶公式：

$$S = \sqrt{\frac{1}{4} \left[ c^2 a^2 - \left( \frac{c^2 + a^2 - b^2}{2} \right)^2 \right]}$$

此與海龍公式等價，顯十三世紀非凡代數造詣。

### A.3 Qin Jiushao’s Formula for Triangular Area

For a triangle with sides  $a, b, c$ , Qin gave the formula:

$$S = \sqrt{\frac{1}{4} \left[ c^2 a^2 - \left( \frac{c^2 + a^2 - b^2}{2} \right)^2 \right]}$$

This is equivalent to Heron’s formula and demonstrates remarkable algebraic insight for the 13th century.

### A.4 盈不足術 / The Surplus and Deficit Method

盈不足術為通用線性問題解法。設兩試值得盈、不足，則真值：

$$x = \frac{a_2 b_1 + a_1 b_2}{a_1 + a_2}$$

其中  $a_1, a_2$  為試入之數， $b_1, b_2$  為對應盈不足之數。

### A.4 The “Surplus and Deficit” Method (Ying Bu Zu)

The method of surplus and deficit (Ying Bu Zu Shu) is a general technique for solving linear problems. Given two trial values yielding surplus and deficit results, the true value is found by:

$$x = \frac{a_2 b_1 + a_1 b_2}{a_1 + a_2}$$

where  $a_1, a_2$  are the trial inputs and  $b_1, b_2$  are the corresponding surplus/deficit amounts. This method can be applied to a wide variety of practical problems.

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*End of Fascicles 3–4 / 卷三至四終*

Shu Xue Jiu Zhang – Juan 3 Shang, Juan 3 Xia, Juan 4 Shang, Juan 4 Xia / 數學九章

# 數學九章

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四庫全書

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問答術草

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三色均價

編者說明

編輯前言

卷五上：賦役 賦役一章，即古九章中差分粟米諸法。但從來算書設題之數未有如此卷之多者。如首題答數至一百七十五條，每條步算之數至十餘位，而得數皆無不合，亦可謂能廣其用矣。且諸問皆足以明貴賤廩稅及交質變易之同異，使公私各得其平，誠治民之要務也。

卷五下：賦役 題語多晦，術草中間有與所問不相應者，亦於各條下指出，俾觀者無惑焉。

卷六上：錢穀 差人五路和糴，課貴賤，折輕齎，皆國家理財之要道。

卷六下：錢穀 推求本息，易牒知原，粟米交易，計米易麴，回運費，三色均價。凡此皆市易之術，亦古粟米章之遺意也。

差分粟米

卷五 賦役類

復邑修賦

問：有海圻縣地，今有復漲，歲以鄉井再成，申諸朮邑。稱土排到六鄉，以附郭為甲，最遠為己，各有田九等，開具下項：

甲鄉共計田十四萬一百九十三畝三角一十二步。上等：上田五千六百七十八畝一角四十八步；中田四千八百九十二畝三十步；下田六千六百二十一畝五十四步。中等：上田八千二百二十五畝二十四步；中田一萬三十五畝六步；下田一萬六千五百三十畝。下等：上田二萬一千九十畝二十四步；中田三萬二千六十畝三步；下田三萬五千六十一畝三角三步。

乙鄉田共計八萬四千一十畝二角二步。丙鄉田共計一十二萬九百三十五畝五十八步五分。丁鄉田共計八萬九千六十六畝二步三分。戊鄉田共計二十萬四千四百七十四畝一角二十四步四分。己鄉田共計一十五萬八千四百六十畝三角一十八步二分。

照得昨來本縣原科苗米一十萬三千五百六十七石八斗四升四合三勺，和買一萬三千四百九十八匹一丈七尺三寸七分六釐，夏稅九千八百七十六匹三丈二尺六寸五分八釐。其六鄉田係三色，甲為上，乙丙為次，丁戊己又為次。先令官物為三差，使上比中、中比下皆十分外差一，次令各鄉九等皆於十分內差一，拋科用合租額。其乙鄉田最肥，次丁，次甲，次丙，次己，次戊。欲知三色九等每畝等則及共科數各鄉幾何？

〔按〕題內言田有三色，甲為上云云，又言乙鄉田最肥云云，語若相左。蓋前言三色者六鄉共科之等，後言田肥者每畝所出之異也。若易田係三色為科有三差，則語意明矣。

答曰：甲鄉：上等上田苗三斗二升三合二勺，和一尺六寸九分，稅一尺二寸三分；中田苗二斗九升九勺，和一尺五寸二分，稅一尺一寸一分；下田苗二斗五升八合六勺，和一尺三寸五分，稅九寸九分。

中等上田苗二斗二升六合二勺，和一尺一寸八分，稅八寸六分；中田苗一斗九升三合九勺，和一尺一分，稅七寸四分；下田苗一斗六升一合六勺，和八寸四分，稅六寸二分。

下等上田苗一斗二升九合三勺，和六寸七分，稅四寸九分；中田苗九升六合九勺，和五寸一分，稅三寸七分；下田苗六升四合六勺，和三寸四分，稅二寸五分。

各鄉共科數：甲鄉苗米一萬九千五百五十石二斗四升八合三勺；和買一萬一百九十二丈二尺六寸五分六釐；夏稅七千四百五十七丈六尺八寸九分八釐。

乙、丙、丁、戊、己各鄉共科數依上法求之。

甲己

苗米和買夏稅

術曰：以衰分求之。先列本縣色位，自下錐行列之，又以鄉數對列而乘之，副併為法，以除官物數，得一分之率。以率數乘未併者，各得諸鄉之數。次列各鄉等位，自上等置十分，每以內分錐行九折之，至九等止，又各以畝步乘之，副併為鄉法，以除諸各鄉所得官物數，所得為一分之率，以乘未併者，各得每畝稅色。

草曰：列本縣色位三，自下色例十分中十一分，上比中身外加一，上得一百二十一，中得一百一十，下得一百，為上中下三率，列之右行。乃列甲一對上率，乙丙共二對中率，丁戊己共三對下率，列左行。乃以左行列數各相對乘右行率數，其上得一百二十一，其十得二百二十，其下得三百，乃副置而併之，得六百四十一為法。

置元科苗米一十萬三千五百六十七石八斗四升四合三勺為寔，以法除之，得一百六十一石五斗七升二合三勺為一分之率。以未併下率一百乘率，得一萬六千一百五十七石二斗三升為丁戊己三鄉各科數。於身下加一，得一萬七千七百七十二石九斗五升三合為乙丙二鄉各科數。又於身下加一，得一萬九千五百五十石二斗四升八合三勺為甲合科數。

求和買亦用六百四十一為法，置元科和買一萬三千四百九十八匹，以匹法四丈通之，得五萬三千九百九十二丈，內零一丈七尺三寸七分六釐，得五萬三千九百九十三丈七尺三寸七分六釐為寔，以法除之，得八十四丈二尺三寸三分六釐為一分之率。亦以未併下率一百乘之，得八千四百二十三丈三尺六寸為丁戊己三鄉各科數。次於身下加一，得九千二百六十五丈六尺九寸六分為乙丙二鄉各科數。又於身下加一，得一萬一百九十二丈二尺六寸五分六釐為甲鄉合科數。

求夏稅亦置六百四十一為法，置元科九千八百七十六匹，以匹法四丈通之，得三萬九千五百四丈，內零三丈二尺六寸五分八釐，得三萬九千五百七丈二尺六寸五分八釐為寔，以法除之，得六十一丈六尺三寸三分八釐為一分率。亦以未併下率一百乘之，得六千一百六十三丈三尺八寸為丁戊己三鄉各科數。次於身下加一，得六千七百七十九丈七尺一寸八分為乙丙二鄉各科數。又於身下加一，得七千四百五十七丈六尺八寸九分八釐為甲鄉數。

衰分



## 圍田租畝

問：有興復圍田已成，共計三千二十一頃五十一畝一十五步，分三等。其上等每畝起租六斗，中等四斗五升，下等四斗。中田多上田弱半，不及下田太半。欲知三色田畝及各租幾何？

〔按〕題言中田多上田弱半，不及下田太半。蓋以弱半為三分之一，太半為三分之二也。多上田弱半者，即比上田多三分之一也。不及下田太半者，即比下田少三分之二也。是上田加三分之一為中田，即下田三分之一也。轉言之，則中田為下田三分之一，上田為中田四分之三也。

答曰：上田四百七十七頃八畝一十五步，米二萬八千六百二十四石八斗三升七合五勺；中田六百三十六頃一十畝三角，米二萬八千六百二十四石八斗三升七合五勺；下田一千九百八頃三十二畝一角，米七萬六千三百三十二石九斗。

術曰：以衰分求之。列母子求田率，副併為法，以共田為寔，寔如法而一，得一分之率。以徧乘未併者，得三等田。各以起租乘之，各得米。

草曰：置弱半母四為中率，子三為上率。以太半子二減母三，餘一，以乘中率四，只得四為中泛。又以餘一乘上率三，只得三為上泛。次以太半母三乘中泛四，得一十二為下泛。副併三泛，得一十九為法。置田三千二十一頃五十一畝，以畝法二百四十通之，得七千二百五十一萬六千二百四十步，內子一十五步，得七千二百五十一萬六千二百五十五步為寔。以法一十九除之，得三百八十一萬六千六百四十五步為一分之數。以上泛三因之，得一千一百四十四萬九千九百三十五步為上積。又以中泛四因之一分數，得一千五百二十六萬六千五百八十步為中積。又以下泛一十二乘一分數，得四千五百七十九萬九千七百四十步為下積。其三積各以畝法二百四十約之為畝。其上田得四百七十七頃八畝一十五步，其中田得六百三十六頃一十畝三角，其下積得一千九百八頃三十二畝一角。各以起租，三積為三寔。其上積一千一百四十四萬九千九百三十五步乘上積六斗，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺為上田租。其中積一千五百二十六萬六千五百八十步乘中田租四斗五升，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺。其下積四千五百七十九萬九千七百四十步乘下租四斗，得一千八百三十一萬九千八百九十六石為寔，以畝法二百四十除之，得七萬六千三百三十二石九斗為下田米。

〔按〕草內求各衰數，意謂以三分為上田衰數，加弱半三分之一得四分為中田衰數，三因中田衰數得一十二分為下田衰數，併之得一十九分為總衰數。其法本屬顯明，但語中參入母子副泛等名目，其意反晦矣。

衰分

×××

戶稅移割

問：某縣據甲稱：本戶田地原納苗三十五石七斗，和買本色一十一匹二丈二尺九寸四分八釐七毫五絲，折帛二十七匹二寸一分三釐七毫五絲，紬絹折帛八匹三丈九寸七寸三分九釐，紬絹本色二十匹二丈八尺九寸三分九釐。已將田四百七畝出與乙，五百十六畝出與丙。訖乞移割本戶所出田上稅賦，歸併乙丙兩戶送納。會到乙原有田三百七十五畝，丙原有田四百六十三畝，並係本鄉本等。每畝苗三升五合，稅一尺一寸五分，物力一貫二百；本等田紬一尺三寸四分，物力九百；物力及三十二貫數和買一匹，內三分本色七折帛；夏稅七分本色三分折帛；紬五分本色五折帛，併絹紬。欲知甲田地及甲乙丙分割合納苗米、和買、夏稅、折帛、本色、畸零各幾何？

〔按〕此題科稅分田地二項。田有科米、稅絹、物力三色，地只有稅紬、物力二色，絹與紬折色不同，物力和買合田地總計之。又甲戶有田者有地，乙丙二戶有田無地。此等皆不可不明言者，題中殊未分晰。

答曰：甲原有田一千二十畝，地一十一畝二角四十八步。  
甲今有田九十七畝，苗米三石三斗九升五合，夏稅折帛三丈三尺四寸六分五釐，本色一匹畸零三丈八尺八分五釐，物力一百一十六貫四百文。地一十一畝二角四十八步，稅紬折帛七尺八寸三分九釐，稅紬畸零七尺八寸三分九釐，物力一十貫五百三十文，田地共物力一百二十六貫九百三十文。和買折帛二匹三丈一尺六分三釐七毫五絲，本色一疋畸零七尺五寸九分八釐七毫五絲。  
乙今有田七百八十二畝，苗米二十七石三斗七升，夏稅折帛六匹二丈九尺七寸九分，本色一十五匹畸零二丈九尺五寸一分，物力九百三十八貫四百文，和買折帛二十匹二丈一尺一寸，本色八匹畸零三丈一尺九寸。  
丙今有田九百七十九畝，苗米三十四石二斗六升五合，夏稅折帛八匹一丈七尺七寸五分五釐，本色一十九匹畸零二丈八尺九分五釐，物力一千一百七十四貫八百文，和買折帛二十五匹二丈七尺九寸五分，本色一十一匹畸零五寸五分。

術曰：以粟米及衰分求之。置甲原納米為實，以每畝苗為法除之，得甲原有田。以乘每畝稅，得田絹。次以甲紬絹本色折帛併之，內減田絹，餘為地紬。以每紬約之，得甲原有地。次以甲出田併乙丙原有田，各得三戶今有田地。各以等則乘之，各得物力苗稅。次以每畝物力率約各戶共物力，得和買。副之，三分因之退位為和本，七分因之退位為和折；夏稅反其分而因之退之；紬以半之，併入稅，各得。

田地

粟米衰分

## 均科綿稅

問：縣科綿有五等戶，共一萬一千三十三戶，共科綿八萬八千三百三十七兩六錢。上等一十二戶，副等八十七戶，中等四百六十四戶，次等二千三十五戶，下等八千四百三十五戶。欲令上三等折半差，下二等比中等六四折差。科率求之，問各戶納及各等幾何？

〔按〕題言下二等比中等六四折差，是次等為中等六分之四，下等又為次等六分之四也。乃術草中皆以十分之四收之，是四分折差而非四六折差矣。集中題與術不相應者多類此，豈成書之後未能重加校正歟？

答曰：上等一戶一百二十四兩，一十二戶計一千四百八十八兩；副等一戶六十二兩，八十七戶計五千三百九十四兩；中等一戶三十一兩，四百六十四戶計一萬四千三百八十四兩；次等一戶一十二兩四錢，二千三十五戶計二萬五千二百三十四兩；下等一戶四兩九錢六分，八千四百三十五戶計四萬一千八百三十七兩六錢。

術曰：列五等戶數，先以四折下等數，加次等戶，又以四折之，加中等戶數，却以半折之，加二等戶，又以半折之，加上等戶數，不折便為法。除科綿，得上等一戶之綿。復半之為副等，又半之為中等，又四折之為次等，又四折之為下等。各一戶綿，却各以戶數乘之，各得五等共出綿。

草曰：先置下等戶八千四百三十五，以四折之，得三千三百七十四。乃以得數折入次戶二千三十五內，得五千四百九戶。又以四折次數五千四百九戶，得二千一百六十三戶六分。乃以得次數併入中戶四百六十四內，共得二千六百二十七戶六分。次以五折中數二千六百二十七戶六分，得數。次以得數一千三百一十三戶八分併副戶八十七，得一千四百戶八分。又以五折副數一千四百戶八分，得七百戶四分，既得數。乃以副數七百戶四分併上戶一十二戶，得七百一十二戶四分，不折便為法。

累折至等，共得七百一十二戶四分以為總法。乃置科綿八萬八千三百三十七兩六錢為實，法七百一十二戶四分而一，得一百二十四兩為上等一戶所出綿。以五折之，得六十二兩為副等一戶所出綿。又五折之，得三十一兩為中等一戶所出綿。乃以四折之，得一十二兩四錢為次等一戶所出綿。又以四折之，得四兩九錢六分為下等一戶所出綿。乃以各等戶數各乘一戶所出，即各得每等共數之綿。

綿折半差六四折

## 均定勸分

問：欲勸糶賑濟，據甲民戶物力畝步排定，共計一百六十二戶，作九等：上等三戶，第二等五戶，第三等七戶，第四等八戶，第五等十三戶，第六等二十一戶，第七等二十六戶，第八等三十四戶，第九等四十五戶。今先觀諭，第一等上戶願糶五千石，第九等戶願糶二百石。欲知各等拋差石數併數認米數各幾何？

答曰：認該米二十三萬七千六百石。

上等一戶米五千石，三戶計一萬五千石；二等一戶米四千四百石，五戶計二萬二千石；三等一戶米三千八百石，七戶計二萬六千六百石；四等一戶米三千二百石，八戶計二萬五千六百石；五等一戶米二千六百石，一十三戶計三萬三千八百石；六等一戶米二千石，二十一戶計四萬二千石；七等一戶米一千四百石，二十六戶計三萬六千四百石；八等一戶米八百石，三十四戶計二萬七千二百石；九等一戶米二百石，四十五戶計九千石。

術曰：以衰分求之。置上下戶米，減餘為實。列等數減一，餘為法。除之，得拋差石數。以差累減上等米，各得諸等米。以各等戶乘之，併之為總數。

草曰：置上等戶米五千石，減下等戶二百石，餘四千八百石為實。以九等減一，餘八為法。除實，得六百石為每等拋差。用減上等，餘四千四百石為二等米。又減六百，得三千八百石為三等米。又減六百，得三千二百石為四等米。又減六百，得二千六百石為五等米。又減六百，得二千石為六等米。又減六百，得一千四百石為七等米。又減六百，得八百石為八等米。又減六百，得二百石為九等米。又各乘戶數，併之，得總認米石數二十三萬七千六百石。

衰分

均定合解

問：州郡合解諸司窠名錢：戶部九十六萬五千四百二十一貫文，總所六十四萬三千六百一十四貫文，運司一萬六千九十貫三百五十文。今諸窠名先催到九千二百五十三貫六百二十文，欲照原額分數均定椿收解，合各幾何？

答曰：戶部五千四百九十七貫二百文；總所三千六百六十四貫八百文；運司九十一貫六百二十文。

術曰：以衰分求之。置諸原率，可約約之。副併為法。以催到錢乘未併者，各為實。實如法而一。

草曰：列戶部九十六萬五千四百二十一貫，總所六十四萬三千六百一十四貫，運司一萬六千九十貫三百五十文，各為原率。今原率可約，求等得一萬六千九十貫三百五十為等數，俱約之：戶部得六十，總所得四十，運司得一，各為率。副併得一百一為法。以催到錢九千二百五十三貫六百二十文乘未併數：六十、四十及一，畢得五十五萬五千二百一十七貫二百文為戶部之實，三十七萬一百四十四貫八百文為總所之實，九千二百五十三貫六百二十文為運司之實。三實皆如法一百一而一：其戶部得五千四百九十七貫二百文，總所得三千六百六十四貫八百文，運司得九十一貫六百二十，各為候解錢。

戶部總所運司

衰分

寬減屯租

問：屯租欲議寬減，仍聽以夏麥折納分數。官牛種者與減二分，私牛種者與減四分。每歲租穀以三分之一許夏折二麥，內四分大六分小折色。每大麥三石折小麥二石，小麥二石折穀三石五斗。屯租舊額官種一石納租五石，私種一石納租三石。今某州屯田去年計官私種共九千七百八十二石，共合收租穀三萬九千五百八十六石。欲知官私種各數、原額、今減、合催、成年夏麥秋穀幾何？

〔按〕此題有官私共種數、租數，有種一石租數，求各種數租數，古謂之差分，今謂之和較比例。折色亦差分也，今謂之和數比例。大小麥與穀相易，古謂之異乘同乘，今謂之和率比列。

答曰：官種五千一百二十石，私出種四千六百六十二石。  
原租額共三萬九千五百八十六石：官種二萬五千六百石，私種一萬三千九百八十六石。  
今減一萬七百一十四石四斗：官種五千一百二十石，私種五千五百九十四石四斗。  
合催二萬八千八百七十一石六斗。  
官種租二萬四百八十石：一分折麥計穀六千八百二十六石六斗六升零三分升之二；四分折大麥二千三百四十石五斗七升七分升之一（係折穀二千七百三十石六斗六升三分升之二）；六分折小麥二千三百四十石五斗七升七分升之一（係折穀四千九十六石二分）；正色穀一萬三千六百五十三石三斗三升三分升之一。  
私種租八千三百九十一石六斗：一分折麥計穀二千七百九十七石二斗；四分折大麥九百五十九石四升（係折穀一千一百一十八石八斗八升）；六分折小麥九百五十九石四升（係折穀一千六百七十八石三斗二升）；二分正色穀五千五百九十四石四斗。  
成年共收：夏折穀九千六百二十三石八斗六升三分升之二；大麥三千二百九十九石六斗一升七分升之一；小麥三千二百九十九石六斗一升七分升之一；秋正穀一萬九千二百四十七石七斗三升三分升之一。

術曰：以衰分求之。置共種數，以官種一石納五石、私種一石納三石為列衰，併之為法。以共租乘未併者為實，實如法而一。各得官私種數。次以各一石租數乘種數，得原額。以減分乘各原額，得今減數。以減原額，餘為合催。次以三分折數乘各催租，得折麥計穀。又於折內分大麥四分、小麥六分，各得麥數。以大麥率三石折小麥二石、小麥二石折穀三石五斗變之，得穀數。以減折麥計穀，餘為正色穀。併之，得成年共收數。

差分折色異乘同乘

衰分

戶稅均寬

問：州郡寬恤，近將某縣下三等稅戶秋科餘欠錢米，已與蠲放：共錢一千三百五十五貫七百六文，米五千二百七十二石一斗九升。某本縣下三等物力計三萬七千六百五十八貫五百文。今來官員陳述：本縣多有樂輸無欠之戶，今蒙蠲放稅尾，似反寬潤頑輸之戶，於理未均。遂議將樂輸三等戶於明年兩稅與照昨來體例減免。契勘得三等無欠戶物力二十二萬八千一十五貫三百二十一文。欲知每百文合減免錢米及共減各幾何？

答曰：每物力一百文，放錢三文六分，放米一升四合。明年兩稅放免：錢七千九百四十九貫三百五十一文五分五釐六毫；米三萬九百一十四石一斗四升四合九勺四抄。

術曰：以粟米衰分求之。置原物力為法，除原放錢米，得每百文物力所放錢米率。以各率乘今來物力，各得錢米，為明年兩稅合放數。

草曰：以原物力三萬七千六百五十八貫五百文為法。先除原放錢一千三百五十五貫七百六文，定法百文上得一文為商，除得三文六分為物力百文所放錢率。次除原放米五千二百七十二石一斗九升，得一升四合為物力百文所放米率。以今三等戶物力二十二萬八千一十五貫三百二十一文徧乘所放錢米二率，得錢七千九百四十九貫三百五十一文五分五釐六毫，得米三萬九百一十四石一斗四升四合九勺四抄，為明年兩稅合放數。

物力

粟米衰分

## 米穀粒分

問：開倉受約，有甲戶米一千五百三十四石。到廊驗得米內夾穀，乃於樣內取米一掬，數計二百五十四粒，內有穀二十八顆。凡粒米率每勺三百。今欲知米內雜穀多少，以折米數科貴及粒各幾何？

答曰：米一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八；穀一百六十九石一斗二合三勺一百二十七分勺之七十九。合折米八十四石五斗五升一合一勺一百二十七分勺之一百三。原米折米共計四十三億四千八百三十四萬六千四百五十六粒。

術曰：以粟米求之，衰分入之。置樣米粒數為法，以常穀顆數減之，餘與穀為列衰。可約約之。以共米乘列衰為各實，實如法而一，各得米數穀數。置穀數以粟率折之，為穀所折米。次以勺率遍乘米數折米，得粒數。

草曰：置一掬樣粒數二百五十四為法，以帶穀二十八顆為穀衰，以減法，餘二百二十六為米衰。此二衰與法皆可約，求等得二，俱以約之：法得一百二十七，米衰得一百一十三，穀衰得一十四。以米一千五百三十四石遍乘二衰，得一十七萬三千三百四十二石為米實，得二萬一千四百七十六石為穀實。皆如法一百二十七而一：米得一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八，穀得一百六十九石一斗二合三勺一百二十七分勺之七十九。以粟率五十折之，得八十四石五斗五升一合一勺一百二十七分勺之一百三為穀折納米數。併二米，得一千四百四十九石四斗四升八合八勺一百二十七分勺之二十四。先通分納子一十八萬四千八十石，以勺率三百粒乘之，得五千五百二十二億四千萬粒為實，以母一百二十七除之，得四十三億四千八百三十四萬六千四百五十六粒，不盡八十八棄之。

穀折米

粟米衰分



卷六 錢穀類

課糴

問：差人五路和糴。據浙西平江府石價三十五貫文，一百三十五合，至鎮江水脚錢每石九百文；安吉州石價二十九貫五百文，一百一十合，至鎮江水脚錢每石一貫二百文；江西隆興府石價二十八貫一百文，一百一十五合，至建康水脚錢每石一貫七百元；吉州石價二十五貫八百五十文，一百二十合，至建康水脚錢每石二貫九百元；湖南潭州石價二十七貫三百文，一百一十八合，至鄂州水脚錢每石一貫七百元。其錢並十七界官會，其米並用文思院斛交量紐數。欲皆以官解計石錢相比貴賤幾何？

〔按〕按草中係二貫一百文，此訛。文思院解每斗八十三合。

答曰：文思院斛石錢：安吉州二十三貫一百六十四文一十一分文之六；平江府二十二貫七十一文二十七分文之二十三；隆興府二十一貫五百七文二十三分文之一十九；潭州二十貫六百七十九文五十九分文之四十九；吉州一十九貫八百八十五文十二分文之五。

〔按〕按三十九訛四十九。

術曰：以粟米互換求之。置石價併水脚，乘石數，又乘官斗合數為實。各如本州合數而一，各得官解石錢。以課貴賤。

草曰：置安吉州石價二十九貫五百文，平江石價三十五貫文，隆興石價二十八貫一百文，吉州石價二十五貫八百五十文，潭州石價二十七貫三百文，列右行。次置水脚：安吉一貫二百文，平江九百元，隆興一貫七百元，吉州二貫九百元，潭州二貫一百文，列左行，各對本州石價。以兩行數併之，得數：安吉三十貫七百元，平江三十五貫九百，隆興二十九貫八百，潭州二十九貫四百，吉州二十八貫七百五十，仍於右行。次以文思院官斗八十三合遍乘之：安吉州得二千五百四十八貫一百文，平江府得二千九百七十九貫七百元，江西隆興得二千四百七十三貫四百文，湖南潭州得二千四百四十貫二百文，江南吉州得二千三百八十六貫二百五十文，各為實於右。得次列安吉斗一百一十合，平江斗一百三十五合，隆興斗一百一十五合，潭州斗一百一十八合，吉州斗一百二十合於左行為法。以對除右行之實：安吉得二十三貫一百六十四文一十一分文之六，平江得二十三貫七十一文二十七分文之二十三，隆興得二十一貫五百七文二十三分文之一十九，潭州得二十貫六百七十九文五十九文分之四十九，吉州得一十九貫八百八十五文一十二分文之五。相課石價，其安吉州最貴，平江次之，隆興又次之，潭州又次之，吉州最賤。

官會

粟米互換

## 折解輕齋

問：有甲乙丙丁四郡，各合起上供銀絹。

甲郡：銀三千二百兩，每兩二貫二百文足；絹六萬四千匹，每匹二貫文足。去京一千里，每擔一里傭錢六文足。其時舊會每貫五十四文足。

乙郡：銀二千七百兩，每兩二貫三百文足；絹四萬九千二百匹，每匹二貫四百二十文足。去京九百八十里，每擔一里傭錢四文二分足。舊會價五十九文足。

丙郡：銀四千兩，每兩新會九貫三百文；絹七萬三千六百匹，每匹新會一十貫三百文。去京二千里，每擔一里傭錢八十文舊會。

丁郡：銀二千六百兩，每兩五十一貫文舊會；絹三萬二千三十五匹，每匹五十八貫文舊會。去京一千五百里，每擔一里傭錢一百文舊會。

諸郡銀每五百兩、絹每六十匹、新會每五千貫為擔。欲並折新會，均作三限起解。求各郡每限及本色原理、折解實用、寬餘、傭錢各新會幾何？

〔按〕此題貫數分三項：其一足數，每千文為一貫；其一舊會數，如甲以五十四文為一貫，乙以五十九文為一貫是也；其一新會數，為舊會數之五倍，如甲以二百七十文為一貫，乙以二百九十五文為一貫是也。四郡或言足數，或言舊會數、新會數。並折新會數。傭錢原以銀五百兩或絹六十匹各為一擔，今皆折新會數，以五千貫為一擔，故有寬餘錢數。題語多未詳，而併為一擔句尤混，略為分析而術草之意大概可見矣。

答曰：甲郡：合解五十萬一百四十八貫一百四十八文。初限一十六萬六千七百一十六貫四十九文，次限一十六萬六千七百一十六貫四十九文，末限一十六萬六千七百一十六貫五十文。傭錢原理二萬三千八百四十五貫九百二十五文二十七分文之二十五，實用二千二百二十二貫八百七十七文二十七分文之二十一，寬餘二萬一千六百二十三貫四十八文二十七分文之四。

乙郡：合解四十二萬四千六百五十七貫六百二十七文。初限一十四萬一千五百五十二貫五百四十二文，次限一十四萬一千五百五十二貫五百四十二文，末限一十四萬一千五百五十二貫五百四十三文。傭錢原理一萬一千五百一十六貫四百二十八文五十九分文之二十八，實用一千一百八十五貫一十文二百九十五分文之五，寬餘一萬三百三十一貫四百一十八文五十九分文之二十七。

丙郡：合解七十九萬五千二百八十貫文。初限二十六萬五千九十三貫三百三十三文，次限二十六萬五千九十三貫三百三十三文，末限二十六萬五千九十三貫三百三十四文。傭錢原理三萬九千五百九貫三百三十三文三分文之一，實用五千八十九貫七百九十二文，寬餘三萬四千四百一十九貫五百四十一文三分文之一。

丁郡：合解三十九萬八千一百二十六貫文。初限一十三萬二千七百八貫六百六十六文，次限一十三萬二千七百八貫六百六十六文，末限一十三萬二千七百八貫六百六十八文。傭錢原理一萬四千一百七十三貫五百文，實用二千三百八十八貫七百五十六文，寬餘一萬一千七百八十四貫七百四十四文。

寬餘足舊會新會  
寬餘

術曰：以均輸求之。置各郡銀絹，乘各價，併之，歸足原，展足為舊會，次以五約舊會為新會，各得合解錢。以限數除之，得每限錢，不盡併歸末限。次置里數，乘每里傭價為率。以率乘原銀及原絹，各為傭實。以每擔銀絹率各為法，實如法而一，不滿者亦為擔，併之為原理傭錢。次以率乘合解錢為實，乃以錢物每擔率為法，實如法而一，各得實用傭錢。以減原理傭錢，餘為寬餘傭錢。

均輸

僦直推原

問：房廊數內，一户日納一百五十六文八分足為準。指揮未曾經減者，減三分；已曾經減三分者，減二分；已曾經減二分者，更減二分。今本户累經減者，欲知原額房錢幾何？

答曰：原額三百五十文。

術曰：以衰分求之。列一十分兩行，各三位；列減分，對減右行。以餘者相乘為法。以左行原列相乘，得納錢為實。實如法而一，得原額錢。

草曰：列一十三位於左行，又列一十分三位於右行。其右上減去初減三分，右中減去次減二分，右下減去更減二分。右行餘七八八，以相乘得四百四十八為法。乃以左行三位一十分相乘，得一千為乘率。以乘見日納錢一百五十六文八分，得一百五十六貫八百文為實。實如法而一，得三百五十文，為本户原額戶錢。

衰分

$$7 \times 8 \times 8 = 448$$
$$10 \times 10 \times 10 = 1,000$$

推求本息

問：三庫息例：萬貫以上一釐，千貫以上二釐五毫，百貫以上三釐。甲庫本四十九萬三千八百貫，乙庫本三十七萬三百貫，丙庫本二十四萬六千八百貫。今三庫共約到息錢二萬五千六百四十四貫二百文。其典率：甲反錐差，乙方錐差，丙蒺藜差。欲知原典三例本息各幾何？

〔按〕此即衰分題也。其差有反錐、方錐、蒺藜之名，蓋以一二三遞減如立錐為反錐，以一四九平方遞加為方錐，以一三六三數遞加為蒺藜。是必古有其名也。至以各差求各本，則因各本原依各差入之也。

答曰：甲庫共納息九千五十三貫文：一釐息二千四百六十九貫文，二釐半息四千一百一十五貫文，三釐息二千四百六十九貫文。  
乙庫共納息一萬五十一貫文：一釐息二百六十四貫五百文，二釐半息二千六百四十五貫文，三釐息七千一百四十一貫五百文。  
丙庫共納息六千五百四十貫二百文：一釐息二百四十六貫八百文，二釐半息一千八百五十一貫文，三釐息四千四百四十二貫四百文。

術曰：置諸庫諸色之差，照釐率為三行。縱併之為約率，橫命之為乘率。以約率各約自庫之本，各得。以遍乘未併乘率，然後各以釐率橫乘之，次以縱併之，為各庫共息。

反錐方錐蒺藜

衰分

草曰：置甲庫反錐差，自下置三二一於右行。次置乙庫方錐差，自上置一四九於中行。次置丙庫蒺藜差，自上置一三六於左行，各為三庫上中下三等乘率。乃縱併甲差三二一，得六為甲約率。縱併乙差一四九，得一十四為乙約率。縱併丙差一三六，得一十為丙約率。直命九位數各為上中下乘率。

乃先以約率各約自庫之本：乃以甲約率六約甲本四十九萬三千八百貫，得八萬二千三百貫為甲得。次以乙約率一十四約乙本三十七萬三百貫，得二萬六千四百五十貫為乙得。次以丙約率一十約丙本二十四萬六千八百貫，得二萬四千六百八十貫為丙得。

以各得乘未併乘率：其甲所得八萬二千三百貫，乘反錐乘率三二一，得二十四萬六千九百貫為上率，得一十六萬四千六百貫為中率，得八萬二千三百貫為下率。其乙所得二萬六千四百五十貫，以乘方錐差一四九，得二萬六千四百五十貫為上率，得一十萬五千八百貫為中率，得二十三萬八千五十貫為下率。其丙所得二萬四千六百八十貫，以乘蒺藜差一三六，得二萬四千六百八十貫為上率，得七萬四千四十貫為中率，得一十四萬八千八十貫為下率。

然後各以息釐數乘各庫三乘：其甲以一釐乘上率二十四萬六千九百貫，得二千四百六十九貫為上息；以二釐五毫乘中率一十六萬四千六百貫，得四千一百一十五貫為中息；以三釐乘下率八萬二千三百貫，得二千四百六十九貫為下息。併上中下三息，得九千五十三貫文為甲庫共息。其乙庫以一釐乘上率二萬六千四百五十貫，得二百六十四貫五百文為上息；以二釐五毫乘中率一十萬五千八百貫，得二千六百四十五貫為中息；以三釐乘下率二十三萬八千五十貫，得七千一百四十一貫五百為下息。併上中下三息，得一萬五十一貫文為乙庫共息。其丙庫以一釐乘上率二萬四千六百八十貫，得二百四十六貫八百為上息；以二釐五毫乘中率七萬四千四十貫，得一千八百五十一貫為中息；以三釐乘下率一十四萬八千八十貫，得四千四百四十二貫四百文為下息。併上中下三息，得六千五百四十貫二百文為丙庫共息。併三庫共息，得二萬五千六百四十四貫二百文為總息。

$$3 + 2 + 1 = 61 + 4 + 9 = 141 + 3 + 6 = 10$$

$$493,800 \div 6 = 82,300370,300 \div 14 = 26,450246,800 \div 10 = 24,680$$

$$82,300 \times 3 = 246,90082,300 \times 2 = 164,60082,300 \times 1 = 82,30026,450 \times 1 = 26,45026,450 \times 4 = 105,80026,450 \times 9 = 238,05024,680 \times 1 = 24,68024,680 \times 3 = 74,04024,680 \times 6 = 148,080$$

$$246,900 \times 0.001 = 2,469164,600 \times 0.0025 = 4,11582,300 \times 0.003 = 2,46926,450 \times 0.001 = 264.5105,800 \times 0.0025 = 2,645238,050 \times 0.003 = 7,141.524,680 \times 0.001 = 246.874,040 \times 0.0025 = 1,851148,080 \times 0.003 = 4,442.4$$

易牒知原

〔按〕按舊本此問無題，今增入。

問：出度牒差人營運：每三道易鹽一十三袋，鹽二袋易布八十四疋，布一十五疋易絹三疋半，絹六疋易銀七兩二錢。今趣到銀九千一百七十二兩八錢。欲知原關度牒道數幾何？

答曰：度牒一百八十道。

術曰：以粟米互乘易法求之。列各數以本色相對，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。

草曰：先以度牒三道乘鹽二袋，得六。以乘布一十五，得九十。又乘絹六疋，得五百四十。乃乘銀九萬一千七百二十八錢（將九千一百七十二兩八錢化為錢），得四千九百五十三萬三千一百二十錢為實。次以鹽一十三袋乘布八十四，得一千九十二。以乘絹三疋五分，得三千八百二十二。乃乘銀七兩二錢（化為七十二錢），得二十七萬五千一百八十四錢為法。除實，得一百八十道為原關度牒。

度牒

粟米互乘易法雁翅

$495,331,200 \div 275,184 = 180$

粟米交易

[按] 按舊本此問無題，今增。

問：菽三升易小麥二升，小麥一升五合易油麻八合，油麻一升二合易粳米一升八合。今將菽一十四石四斗欲易油麻，又將小麥二十一石六斗欲易粳米。問各幾何？

答曰：油麻五石一斗二升；粳米一十七石二斗八升。

術曰：以粟米換易求之。置原易率，本色對列，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。各得或問數，不干其率者不置。

草曰：置四色六數，列六位率，如鴈翅，皆化為合。先將菽一十四石四斗化作一萬四千四百合，乃對前二句率數四位，如鴈翅，至欲易油麻止，共五事為上圖。次將小麥二十一石六斗化為二萬一千六百合，乃對後兩句率四位，鴈翅，至欲粳米止，共五事為下圖。

上圖：以菽一萬四千四百合乘麥二十（二升化為二十合），得二十八萬八千，又乘油麻八合，得二百三十萬四千合為油麻實。次以菽三十合（三升化）乘麥一十五合，得四百五十合為法。除之，得五千一百二十合，展為五石一斗二升為油麻。

下圖：以小麥二萬一千六百合乘油麻八合，得一十七萬二千八百合，又乘粳米一十八合，得三百一十一萬四百合為粳米實。以小麥一升五合（十五合）乘油麻一十二合，得一百八十合為法。除之，得一萬七千二百八十合，展作一十七石二斗八升為粳米。

粟米換易

$$\times \times \times 2,304,000 \div 450 = 5,120$$

$$\times \times \times 3,114,000 \div 180 = 17,280$$



## 計米易麴

〔按〕按舊本此問無題，今增。

問：庫率：粳穀七石出米三石，糯米一斗易小麥一斗七升，小麥五升踏麴二斤四兩，麴一百一十斤醖糯米一石三斗。今有糯穀一千七百五十九石三斗八升，欲出穀做米，易麥踏麴，還自醖，餘穀之米須令適足。各合幾何？

答曰：共穀一千七百五十九石三斗八升。出穀九百二十四石，得米三百九十六石。易麥六百七十三石二斗。踏麴三萬二百九十四斤。餘穀八百三十五石三斗八升。醖米三百五十八石二升。

術曰：以粟米換易求之。置諸率，隨本色對列，如鴈翅。有分者通之，異類者變之，以頭位者進乘之，以下位退乘之，得合數。有對者相乘之，無對者直命之，為諸率。併上下無對者為法率。以今有物徧乘諸率（不乘法率），各為實。諸實並如法而一，各得其已變者，復互易乘除之，即得所求。

粟米換易

## 回運費

問：有江西水運米一十二萬三千四百石，原係鎮江交卸，計水程二千一百三十里，每石水脚錢一貫二百文。今截上件米就池州安頓，池州至鎮江八百八十里。欲收回不該水脚錢幾何？

答曰：收回錢六萬一千一百七十八貫五百九十一文。

術曰：以粟米互易求之。置池州至鎮江里數，乘水脚錢，得數又乘運米為實。以原至鎮江水程為法，除實，得收回錢。

草曰：置池州至鎮江八百八十里，乘每石水脚錢一貫二百，得一千五十六貫文。又乘運米一十二萬三千四百石，得一億三千三十一萬四百貫文為實。以原至鎮江水程二千一百三十里為法，除實，得六萬一千一百七十八貫五百九十一文為收回錢。

粟米互易

三色均價

問：庫有三色金共五千兩：內八分金一千二百五十兩，兩價四百貫文；七分五釐金一千六百兩，兩價三百七十五貫文；八分五釐金二千一百五十兩，兩價四百二十五貫文。並欲煉為足色，每兩工食藥炭錢三貫文，耗金九百七十二兩五錢。欲知色分及兩價各幛何？

答曰：色十分；兩價五百三貫七百二十四文，五百三十七分文之二百一十二。

術曰：以方田及粟米求之。置共數，以耗減之，餘為法。以三色分數各乘兩數，併之為色分實。以三色價數各乘兩數為寄，以工藥價乘共金，併寄，共為價實。二實皆如法而一，即各得。

草曰：置共金五千兩，減耗九百七十二兩五錢，外餘四千二十七兩五錢為法。次置一千二百五十兩，乘八分，得一萬分於上。置一千六百兩，以七分五釐乘之，得一萬二千分，加上。置二千一百五十兩，乘八分五釐，得一萬八千二百七十五分，又加上，共得四萬二千七十五分為分實。  
次置一千二百五十兩乘四百貫，得五十萬貫為寄。次置一千六百兩乘價三百七十五貫，得六十萬貫，加寄。次置二千一百五十兩乘價四百二十五貫，得九十一萬三千七百五十貫，又加寄。次置共金五千兩，乘工藥錢三貫，得一萬五千貫，又加寄，共得二百二萬八千七百五十貫為價實。二實並如法四千二十七兩五錢而一：得其色一十分；其價每兩得五百三貫七百二十四文，不盡一貫五百九十。與法求等，得七十五，俱約之，為五百三十七文之二百一十二。

八分足色耗金

方田×

$$\begin{aligned} 1,250 \times 0.8 &= 10,000 \\ 1,600 \times 0.75 &= 12,000 \\ 2,150 \times 0.85 &= 18,275 \\ 40,275 \div 4,027.5 &= 10 \\ 1,250 \times 400 &= 500,000 \\ 1,600 \times 375 &= 600,000 \\ 2,150 \times 425 &= 913,750 \\ 500,000 + 600,000 + 913,750 &= 1,013,750 \\ 1,013,750 \div 2,027.5 &= 500 \end{aligned}$$

編輯說明

一、文本來源

本文以四庫全書本數學九章為底本，參校永樂大典殘卷及宜稼堂叢書本。四庫本雖經館臣整理，然訛誤脫漏之處尚多，題與術不相應者尤為習見。故編輯者於各條下加按語，以明其得失。

二、術語對照

本文採用雙欄對照形式，左欄為原文，右欄為英譯。問答術草四部結構一仍其舊，按語以方括號標出。度量衡名詞保留音譯，加以解釋；數學術語則酌用現代英語對應詞，並附拉丁轉寫。

三、算理說明

卷五賦役類多涉及差分之法，即今之所謂比例分配也。卷六錢穀類多涉及粟米換易，即今之所謂連鎖比例也。秦九韶於古術多所推廣，數位繁曠而條理分明，可謂造詣精深矣。

四、校勘記

本文校勘以四庫全書本為主，凡有訛誤者，編輯按語中隨文指出。如課糴題「三十九訛四十九」之類，皆據前後術草推算所得，非臆斷也。

五、參考文獻

秦九韶原著，四庫全書本數學九章

錢寶琮主編，中國數學史，

李儼，中國古代數學史料，

問答術草

衰分粟米換易

# 數學九章

雙語版  
卷七至卷九

作者 (宋) 秦九韶撰  
分類 營建類 · 軍旅類 · 市易類

|     |     |
|-----|-----|
| 卷七上 | 營建類 |
| 卷七下 | 營建類 |
| 卷八上 | 軍旅類 |
| 卷八下 | 軍旅類 |
| 卷九上 | 市易類 |
| 卷九下 | 市易類 |

導言

卷七上

計作清臺

計築<sup>㊦</sup>岸

計開河渠

卷七下

計作石坝

計修城池

計造浮橋

計開倉窖

卷八上

計立方營

計圓營

計望敵<sup>㊦</sup>

卷八下

計軍衣布

計造軍衣

計載糧運

計營糧

卷九上

計<sup>㊦</sup>米粟

計均貨值

計求美茶

卷九下

計求鹽價

計均錢值

計定物價

計販運得利

編輯附<sup>㊦</sup>

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## 導言數學九章秦九韶九章算術

大衍類

天時類

田域類

測望類

賦役類

錢<sub>口</sub>類

營建類

軍旅類

市易類

淡<sub>口</sub>底色淡藍底色問→答→術→草

卷七上

營建類

卷七營建類包含建築工程、土方計算、水利工程及結構規劃等問題。此類問題涉及計算體積、材料需求、勞動力分配及實際建築工程的幾何尺寸。

**計作清臺問：**問築清臺一所，正高一十二丈，上廣五丈，袤七丈，下廣一十五丈，袤一十七丈。其袤當東西，廣當南北。北秋程人日行六十里，里法三百六十步。土鍤土每工各二百尺，築土每工九十尺，每擔土壤一丈遠。今欲築此臺，問：需要多少工日？

**答曰：**開方及積尺數，以定工日。用工若干萬工。  
**術曰：**按臺體方塹堵，用上廣袤、下廣袤及高相求，得積數。臺積術曰：上下廣袤各自相乘，又上下廣袤交叉相乘，方之，以高乘之，三而一，得臺積。以各工率除之，得工日。

上廣= 5丈，上袤= 7丈，下廣= 15丈，下袤= 17丈，高= 12丈。

上廣×上袤= 5 × 7 = 35

下廣×下袤= 15 × 17 = 255

上廣×下袤= 5 × 17 = 85

上袤×下廣= 7 × 15 = 105

四數相并：35 + 255 + 85 + 105 = 480

以高乘之：480 × 12 = 5760

三而一： $\frac{5760}{3} = 1920$ （立方丈）

方立方尺：1920 × 1000 = 1,920,000（立方尺）

按各工率均攤，得總工若干。

塹堵

= 5 = 7 = 15 = 17 = 12

× = 5 × 7 = 35

× = 15 × 17 = 255

× = 5 × 17 = 85

× = 7 × 15 = 105

35 + 255 + 85 + 105 = 480

480 × 12 = 5760

$\frac{5760}{3} = 1920$

$\frac{3}{1920} \times 1000 = 1,920,000$

**計築方岸問：**問築方一道，長一千八百丈。其方截面：上闊一丈，下闊三丈，高一丈二尺。問：積土若干？用夫若干？

**答曰：**積土二百八十八萬立方尺，用夫若干。  
**術曰：**方體方長條截頭體。以上下闊并之，半之，得平均闊；以高乘之，得截面積；以長乘之，得積。  
**術曰：**上下廣各一，并而半之，以高乘之，又以長乘之，得堤積。

上闊= 1丈，下闊= 3丈，高= 1.2丈，長= 1800丈。

上下闊并半： $\frac{1+3}{2} = 2$ （平均闊）

截面積：2 × 1.2 = 2.4（平方丈）

= 1 = 3 = 1.2 = 1800

$\frac{1+3}{2} = 2$

2 × 1.2 = 2.4

2.4 × 1800 = 4320



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積土： $2.4 \times 1800 = 4320$ （立方丈）  
[尺]： $4320 \times 1000 = 4,320,000$ （立方尺）

$4320 \times 1000 = 4,320,000$

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**計開河渠問：**問開河渠一道，長三千六百丈。渠口上廣八丈，底廣四丈，深二丈。每工日穿土六十尺，負土一尺遠。問：積土及用工各若干？  
**答曰：**積土若干立方尺，用工若干萬工。  
**術曰：**渠體[尺]塹堵。上下廣并之，半之，以深乘之，得截面積；又以長乘之，得積。術同堤法。

上廣= 8丈，底廣= 4丈，深= 2丈，長= 3600丈。

上下廣并半： $\frac{8+4}{2} = 6$ （平均闊）

截面積： $6 \times 2 = 12$ （平方丈）

積土： $12 \times 3600 = 43200$ （立方丈）

[尺]： $43200 \times 1000 = 43,200,000$ （立方尺）

用工： $\frac{43200000}{60} = 720,000$ （工日）

$$\begin{aligned} &= 8+ 4= 2= 3600 \\ &\frac{8+4}{2} = 6 \\ &6 \times 2 = 12 \\ &12 \times 3600 = 43,200 \\ &43200 \times 1000 = 43,200,000 \\ &\frac{43,200,000}{60} = 720,000 \end{aligned}$$

## 卷七下

### 營建類（續）

**計作石坝問：**問作石坝一座，長五十丈，高一十丈，迎水面斜長十二丈，背水面斜長八丈。頂闊三丈，底闊一十丈。問：積石若干？

**答曰：**積石若干立方丈。

**術曰：**坝體截面 $\square$ 梯形。以頂闊、底闊并之，半之，以高乘之，得截面積；以長乘之，得積。

頂闊= 3丈，底闊= 10丈，高= 10丈，長= 50丈。

$$\text{上下并半：} \frac{3+10}{2} = 6.5$$

$$\text{截面積：} 6.5 \times 10 = 65 \text{（平方丈）}$$

$$\text{積石：} 65 \times 50 = 3250 \text{（立方丈）}$$

$$\begin{aligned} &= \frac{3+10}{2} \times 10 = 65 \\ &65 \times 50 = 3250 \end{aligned}$$

**計修城池問：**問修城一段，城周回一千二百丈。其城截面：頂闊二丈五尺，底闊八丈，高三丈。又有女 $\square$ 二十四座，每座長二丈，闊一丈，高八尺。問：共積土若干？

**答曰：**城積及女 $\square$ 積共若干立方丈。

**術曰：**城體 $\square$ 環形截頭體，以周回 $\square$ 長。術曰：頂底闊并半，以高乘之，得截面積；以城周回乘之，得城積。女 $\square$ 各 $\square$ 長方體，長寬高相乘得積，以座數乘之。并城 $\square$ 與女 $\square$ 積，得總數。

城截面：頂闊= 2.5丈，底闊= 8丈，高= 3丈，周= 1200丈。

$$\text{上下并半：} \frac{2.5+8}{2} = 5.25$$

$$\text{城截面積：} 5.25 \times 3 = 15.75 \text{（平方丈）}$$

$$\text{城積：} 15.75 \times 1200 = 18900 \text{（立方丈）}$$

$$\text{女}\square\text{積：} 2 \times 1 \times 0.8 = 1.6 \text{（立方丈座）}$$

$$\text{女}\square\text{總積：} 1.6 \times 24 = 38.4 \text{（立方丈）}$$

$$\text{總積：} 18900 + 38.4 = 18938.4 \text{（立方丈）}$$

女 $\square$

$$\begin{aligned} &= \frac{2.5+8}{2} \times 3 = 15.75 \\ &15.75 \times 1200 = 18,900 \\ &2 \times 1 \times 0.8 = 1.6 \\ &1.6 \times 24 = 38.4 \\ &18900 + 38.4 = 18,938.4 \end{aligned}$$

**計造浮橋問：**問造浮橋一所以濟師。河闊三百丈，每舟長五丈，闊一丈二尺，深八尺。舟上鋪板，板厚六寸。問：需舟若干？板積若干？

**答曰：**需舟若干艘，板積若干立方丈。

**術曰：**以河闊除以舟長，得舟數。以舟面積乘板厚，得板積。

河闊= 300丈，舟長= 5丈，舟闊= 1.2丈。

$$\text{舟數：} \frac{300}{5} = 60 \text{（艘）}$$

$$\text{每舟面積：} 5 \times 1.2 = 6 \text{（平方丈）}$$

$$\text{板積：} 6 \times 0.06 \times 60 = 21.6 \text{（立方丈）}$$

$$\begin{aligned} &= \frac{300}{5} = 60 \\ &5 \times 1.2 = 6 \\ &6 \times 0.06 \times 60 = 21.6 \end{aligned}$$

**計開倉窖問：**問開倉窖一所，上口方二丈，底方一丈二尺，深一丈八尺。問：容積及積土各若干？

答曰：容積若干立方丈，積土若干立方丈。  
術曰：倉窖方錐臺。術曰：上下方各自乘，上下方相乘，并之，以深乘之，三而一。

上方= 2丈，下方= 1.2丈，深= 1.8丈。

上方自乘：  $2 \times 2 = 4$

下方自乘：  $1.2 \times 1.2 = 1.44$

上下方相乘：  $2 \times 1.2 = 2.4$

三數并：  $4 + 1.44 + 2.4 = 7.84$

以深乘：  $7.84 \times 1.8 = 14.112$

三而一：  $\frac{14.112}{3} = 4.704$  (立方丈)

$$= 2 = 1.2 = 1.8$$

$$2 \times 2 = 4$$

$$1.2 \times 1.2 = 1.44$$

$$2 \times 1.2 = 2.4$$

$$4 + 1.44 + 2.4 = 7.84$$

$$7.84 \times 1.8 = 14.112$$

$$\frac{14.112}{3} = 4.704$$

## 卷八上

### 軍旅類

卷八軍旅類涉及軍事後勤問題：營地 $\square$ 局、陣形幾何、估算敵軍數量、計算軍衣糧 $\square$ 供應，以及規劃糧草與軍需品的運輸。這些問題對於南宋抵抗北方入侵者的國防事業具有極重要的實用意義。

**計立方營問：**問一軍三將，將三十三隊，隊一百二十五人。遇暮立營，人占地八尺。須令隊間容隊，師居中央。欲知營方幾何？

**答曰：**方營一百七十一丈，隊方九丈。

**術曰：**先計總人數，以每人占地八尺乘之，得營總積。乃開平方，得營方。又以每隊人數乘占地，開平方，得隊方。營方減隊方而半之，得隊距。

總隊數： $3 \times 33 = 99$ （隊）

總人數： $99 \times 125 = 12375$ （人）

營總積： $12375 \times 8 = 99000$ （平方尺）

$\square$ 平方丈： $\frac{99000}{100} = 990$ （平方丈）

開平方得營方： $\sqrt{990} \approx 31.5$ （丈）按古法修正 $\square$ 整數。

按三三隊布陣，營方：171（丈），隊方：9（丈）。

$$3 \times 33 = 99$$

$$99 \times 125 = 12,375$$

$$12,375 \times 8 = 99,000$$

$$\frac{99,000}{100} = 990$$

$$\sqrt{990} \approx 31.5$$

$$1719$$

**計圓營問：**問立圓營一匝，馬軍五千騎，步軍三萬人。騎兵占地二丈四尺，步兵占地九尺。令步居 $\square$ ，騎居外，各 $\square$ 方陣。問：營圓徑幾何？

**答曰：**圓營徑若干丈。

**術曰：**以人數各乘占地，得步兵積、騎兵積。各開平方，得 $\square$ 外方邊。以 $\square$ 方加兩倍騎陣厚，得外方。乃以方求圓，得營徑。

步兵積： $30000 \times 0.9 = 27000$ （平方丈）

步兵方邊： $\sqrt{27000} \approx 164.3$ （丈）

騎兵積： $5000 \times 2.4 = 12000$ （平方丈）

騎兵方邊： $\sqrt{12000} \approx 109.5$ （丈）

按圓徑術求之，得營圓徑若干丈。

$$30,000 \times 0.9 = 27,000$$

$$\sqrt{27,000} \approx 164.3$$

$$5000 \times 2.4 = 12,000$$

$$\sqrt{12,000} \approx 109.5$$

**計望敵 $\square$ 問：**問敵軍臨河下寨。於我岸高山上，立一表高三丈。 $\square$ 望敵營，表末與敵人目及。退行一十丈，伏地望之，表末與敵人目又及。問：敵營去表及敵人各若干？敵 $\square$ 約幾何？

**答曰：**敵營去表若干丈，敵人目高若干尺，敵 $\square$ 約若干。

**術曰：**此 $\square$ 重差之術。以表高乘退行 $\square$ 實，以兩望之差 $\square$ 法，實如法而一，得敵營去表之遠。又以表高乘表去敵營，以退行除之，得敵人目高。以營地積及人占推之，得敵 $\square$ 約數。

表

卷八下

軍旅類（續）

**計軍衣布問：**問今有軍三萬二千四百人，每人給 ㊦襖被 ㊦一腰，用布七尺五寸；給襖一領，用布一丈二尺；給被一條，用布二丈四尺。問：共布幾何？

**答曰：**共布若干匹（每匹四丈）。  
**術曰：**以人數乘每人用布，得總布數。術曰：各以人數乘本用布，并而㊦總。以匹法四丈除之，得匹數。

|                                       |                                  |
|---------------------------------------|----------------------------------|
| 每人用布：7.5 + 12 + 24 = 43.5（尺）          | $7.5 + 12 + 24 = 43.5$           |
| 總布：32400 × 43.5 = 1,409,400（尺）        | $32,400 \times 43.5 = 1,409,400$ |
| ㊦匹： $\frac{1409400}{40} = 35,235$ （匹） | $\frac{1,409,400}{40} = 35,235$  |

**計造軍衣問：**問造軍衣，每人一襲。每襲用布一丈四尺，裏布七尺， ㊦一兩二錢，棉一斤八兩。今有軍一萬八千人，問：物料各若干？

**答曰：**布若干匹，裏布若干匹， ㊦若干斤，棉若干斤。  
**術曰：**以軍人數各乘每襲所用物料，得總數。以各法約之。

|                                             |                                       |
|---------------------------------------------|---------------------------------------|
| 面布總：18000 × 14 = 252,000（尺） = 6300（匹）       | $18,000 \times 14 = 252,000 = 6300$   |
| 裏布總：18000 × 7 = 126,000（尺） = 3150（匹）        | $18,000 \times 7 = 126,000 = 3150$    |
| ㊦總：18000 × 1.2 = 21,600（錢） = 135（斤）         | $18,000 \times 1.2 = 21,600 = 135$    |
| 棉總：每人棉 = 1斤8兩 = 1.5斤，18000 × 1.5 = 27000（斤） | $= 1 = 1.518,000 \times 1.5 = 27,000$ |

**計載糧運問：**問運糧十萬石，每車載五十石。車行日行六十里，空車日行八十里。裝卸各一日。問：車若干輛？日若干？

**答曰：**車若干輛，往返若干日。  
**術曰：**以總糧除以每車載數，得車數。以行程及空重車速，求往返日數。術曰：總糧 ㊦實，每車載 ㊦法，實如法而一，得車數。又以里數求空重日行，并裝卸日，得往返總日。

|                                    |                             |
|------------------------------------|-----------------------------|
| 車數： $\frac{100000}{50} = 2000$ （輛） | $\frac{100,000}{50} = 2000$ |
| 裝車日：1日，卸車日：1日                      |                             |
| 重車日：按行程若干里，重車行60里日                 |                             |
| 空車日：同行程，空車行80里日                    |                             |
| 往返總日 = 1 + 1 + 重車日 + 空車日           | $= 1 + 1 + +$               |

**計營糧問：**問一軍二萬五千人，每人日食米二升。今有米三萬石，問：足幾日食？

**答曰：**足若干日食。  
**術曰：**以人數乘日食量，得日食總數。以總米除以日食數，得足日。

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日食總：  $25000 \times 2 = 50000$  (升) = 500 (石)

足日：  $\frac{30000}{500} = 60$  (日)

$25,000 \times 2 = 50,000 = 500$

$\frac{30,000}{500} = 60$

## 卷九上

### 市易類

卷九市易類處理商業、 $\square$ 收、貨幣 $\square\square$ 、定價及市場交易等問題。這些問題反映了南宋（）高度發達的商業經濟，及其 $\square$ 雜的貨幣體系、商品貿易，以及政府對鹽和茶的專賣制度。

**計 $\square$ 米粟問：**問 $\square$ 米三千石，每石價錢二貫五百文。今持銀一錠，重五十兩，每兩折錢一貫二百文。問：銀足否？余欠若干？

**答曰：**銀不足，欠若干貫。

**術曰：**以米數乘石價，得總價。以銀重乘每兩折錢，得銀值。以銀值減總價，得余欠。

總價：3000  $\times$  2500 = 7,500,000（文）= 7500（貫）

銀值：50  $\times$  1200 = 60,000（文）= 60（貫）

余欠：7500 - 60 = 7440（貫）

$$3000 \times 2500 = 7,500,000 = 7500$$

$$50 \times 1200 = 60,000 = 60$$

$$7500 - 60 = 7440$$

**計均貨值問：**問甲有絹三百匹，每匹價三貫；乙有布五百匹，每匹價一貫二百文；丙有 $\square$ 二百匹，每匹價五貫。三人欲共買一船貨，價一萬貫。問：各出幾何？

**答曰：**甲出若干貫，乙出若干貫，丙出若干貫。

**術曰：**此 $\square$ 均輸之術。以各人所持貨值 $\square$ 率，按率分配總價。術曰：各以匹數乘本價，得總值 $\square$ 列衰。并以 $\square$ 法，以總價乘列衰，實如法而一，各得所出。

甲值：300  $\times$  3 = 900（貫）

乙值：500  $\times$  1.2 = 600（貫）

丙值：200  $\times$  5 = 1000（貫）

總值：900 + 600 + 1000 = 2500（貫）

甲出： $\frac{900}{2500} \times 10000 = 3600$ （貫）

乙出： $\frac{600}{2500} \times 10000 = 2400$ （貫）

丙出： $\frac{1000}{2500} \times 10000 = 4000$ （貫）

絹布 $\square$

$$300 \times 3 = 900$$

$$500 \times 1.2 = 600$$

$$200 \times 5 = 1000$$

$$900 + 600 + 1000 = 2500$$

$$\frac{900}{2500} \times 10,000 = 3600$$

$$\frac{600}{2500} \times 10,000 = 2400$$

$$\frac{1000}{2500} \times 10,000 = 4000$$

**計求美茶問：**問官買茶三處：甲處茶八千斤，每斤價三百二十文；乙處茶一萬二千斤，每斤價二百八十文；丙處茶一萬五千斤，每斤價二百五十文。欲均 $\square$ 一價發賣，問：每斤均價幾何？

**答曰：**均價若干文。

**術曰：**此 $\square$ 均價之術。以各處斤數乘本價，得各處總價；并之 $\square$ 實；并各處斤數 $\square$ 法；實如法而一，得均價。

甲總價：8000  $\times$  320 = 2,560,000（文）

乙總價：12000  $\times$  280 = 3,360,000（文）

$$8000 \times 320 = 2,560,000$$

$$12,000 \times 280 = 3,360,000$$

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丙總價：  $15000 \times 250 = 3,750,000$  (文)  
總價：  $2560000 + 3360000 + 3750000 = 9,670,000$   
(文)  
總斤：  $8000 + 12000 + 15000 = 35000$  (斤)  
均價：  $\frac{9670000}{35000} \approx 276.3$  (文斤)

$15,000 \times 250 = 3,750,000$   
 $2,560,000 + 3,360,000 + 3,750,000 = 9,670,000$   
 $8000 + 12,000 + 15,000 = 35,000$   
 $\frac{9,670,000}{35,000} \approx 276.3$



## 卷九下

### 市易類（續）

**計求鹽價問：**問官鬻鹽，每袋重三百斤，成本一百貫。欲得利三成，問：每袋賣價幾何？每斤賣價幾何？

**答曰：**每袋賣價若干貫，每斤若干文。

**術曰：**以成本乘利加成本，得賣價。術曰：以本 $\square$ 一，加三成 $\square$ 一三，以本乘之，得袋價。以袋價除以斤數，得斤價。

$$\text{成本} = 100 \text{ 貫，利} = 30$$

$$\text{袋價} = 100 \times 1.3 = 130 \text{ (貫)}$$

$$\text{斤價} = \frac{130000}{300} \approx 433.3 \text{ (文斤)}$$

$$= 100 = 30$$

$$100 \times 1.3 = 130$$

$$\frac{130,000}{300} \approx 433.3$$

**計均錢值問：**問庫中舊錢三種：一曰足陌錢，陌法一千文；二曰九陌錢，陌法九百文；三曰八陌錢，陌法八百文。今有足陌錢五百貫，九陌錢三百貫，八陌錢二百貫。欲均 $\square$ 一陌，問：合一千文 $\square$ 陌，共得若干貫？

**答曰：**共得若干貫（足陌）。

**術曰：**以各陌錢數乘本陌法，得實際文數；并之；以足陌法一千除之，得共數。

$$\text{足陌實數} = 500 \times 1000 = 500,000 \text{ (文)}$$

$$\text{九陌實數} = 300 \times 900 = 270,000 \text{ (文)}$$

$$\text{八陌實數} = 200 \times 800 = 160,000 \text{ (文)}$$

$$\text{總實數} = 500000 + 270000 + 160000 = 930,000 \text{ (文)}$$

$$\text{合足陌} = \frac{930000}{1000} = 930 \text{ (貫)}$$

足陌錢九陌錢八陌錢

$$500 \times 1000 = 500,000$$

$$300 \times 900 = 270,000$$

$$200 \times 800 = 160,000$$

$$500,000 + 270,000 + 160,000 = 930,000$$

$$\frac{930,000}{1000} = 930$$

**計定物價問：**問有絲、綿、麻三物，共值八百貫。絲一斤價四貫，綿一斤價二貫，麻一斤價五百文。欲買絲、綿、麻各若干斤，使價相等。問：各得幾何？

**答曰：**絲若干斤，綿若干斤，麻若干斤。

**術曰：**以總價三而一，得每物之值。各以本價除之，得斤數。

$$\text{每物之值} = \frac{800}{3} \approx 266.67 \text{ (貫)}$$

$$\text{絲斤數} = \frac{266.67}{4} \approx 66.67 \text{ (斤)}$$

$$\text{綿斤數} = \frac{266.67}{2} \approx 133.33 \text{ (斤)}$$

$$\text{麻斤數} = \frac{266.67}{0.5} = 533.33 \text{ (斤)}$$

絲綿麻

$$\frac{800}{3} \approx 266.67$$

$$\frac{266.67}{4} \approx 66.67$$

$$\frac{266.67}{2} \approx 133.33$$

$$\frac{266.67}{0.5} = 533.33$$

**計販運得利問：**問一人持絹五百匹至遠方販賣。去時每匹價四貫，到彼時價漲五成。回程買茶，茶價每斤三百文。問：賣絹得錢若干？可買茶若干斤？

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**答曰：**得錢若干貫，可買茶若干斤。  
**術曰：**以本價加利，得賣價；以匹數乘之，得總錢。  
以總錢除以茶價，得茶斤數。

賣價： $4 \times 1.5 = 6$ （貫匹）

總錢： $500 \times 6 = 3000$ （貫） $= 3,000,000$ （文）

茶斤數： $\frac{3000000}{300} = 10000$ （斤）

$$4 \times 1.5 = 6$$

$$500 \times 6 = 3000 = 3,000,000$$

$$\frac{3,000,000}{300} = 10,000$$

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編輯附[F]  
度量衡單位本卷各題[F]用傳統中國度量衡單位，  
[F]列對照表如下：

| 中文   |         |
|------|---------|
| 長度單位 |         |
| 丈    | ≈ 3.33  |
| 尺    | ≈ 33.3  |
| 寸    | ≈ 3.33  |
| 面積單位 |         |
| 平方丈  | ≈ 11.09 |
| 體積單位 |         |
| 立方丈  | ≈ 37.0  |
| 立方尺  | ≈ 37.0  |
| 容量單位 |         |
| 石    | ≈ 60    |
| 升    | ≈ 0.6   |
| 重量單位 |         |
| 斤    | ≈ 600   |
| 兩    | ≈ 37.5  |
| 錢    | ≈ 3.75  |
| 貨幣單位 |         |
| 貫    | 銅錢      |
| 文    | 一枚銅錢    |
| 陌    |         |
| 里程單位 |         |
| 里    | ≈ 556   |
| 步    | ≈ 1.54  |

數學方法《數學九章》運用了若干具有代表性的  
數學技術：

一、臺體體積公式

對於上廣 $a$ 、上袤 $b$ 、下廣 $c$ 、下袤 $d$ 、高 $h$ 之長方截  
頭臺（塹堵），其體積[F]：

$$V = \frac{h}{3}(ab + cd + ad + bc)$$

$$a \times bc \times dh$$
$$V = \frac{h}{3}(ab + cd + ad + bc)$$

此即現代稱[F]擬柱體公式之特例。

二、衰分術（比例分配）

衰分

按給定比率進行分配的問題，以各人之率[F]列衰，  
并[F]法，以總數乘列衰，實如法而一。

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### 三、重差術

利用兩次觀測以測定遠距離與高度的三角測量技法。

### 四、陌法與算

南宋時期銅錢貶值，出現各種陌法（九陌、八陌、七陌等），反映當時複雜的貨幣行情。

**歷史背景** 秦九韶於年完成《數學九章》，時值南宋末年，國勢動盪。卷八軍旅類中的問題直接反映了當時緊迫的軍事形勢——宋朝面臨蒙古大軍（成吉思汗及其後繼者忽必烈）的存亡威脅。

卷九市易類則反映了宋代發達的商業經濟——包括紙幣流通、政府對鹽茶的專賣制度，以及廣泛的長途貿易網絡。